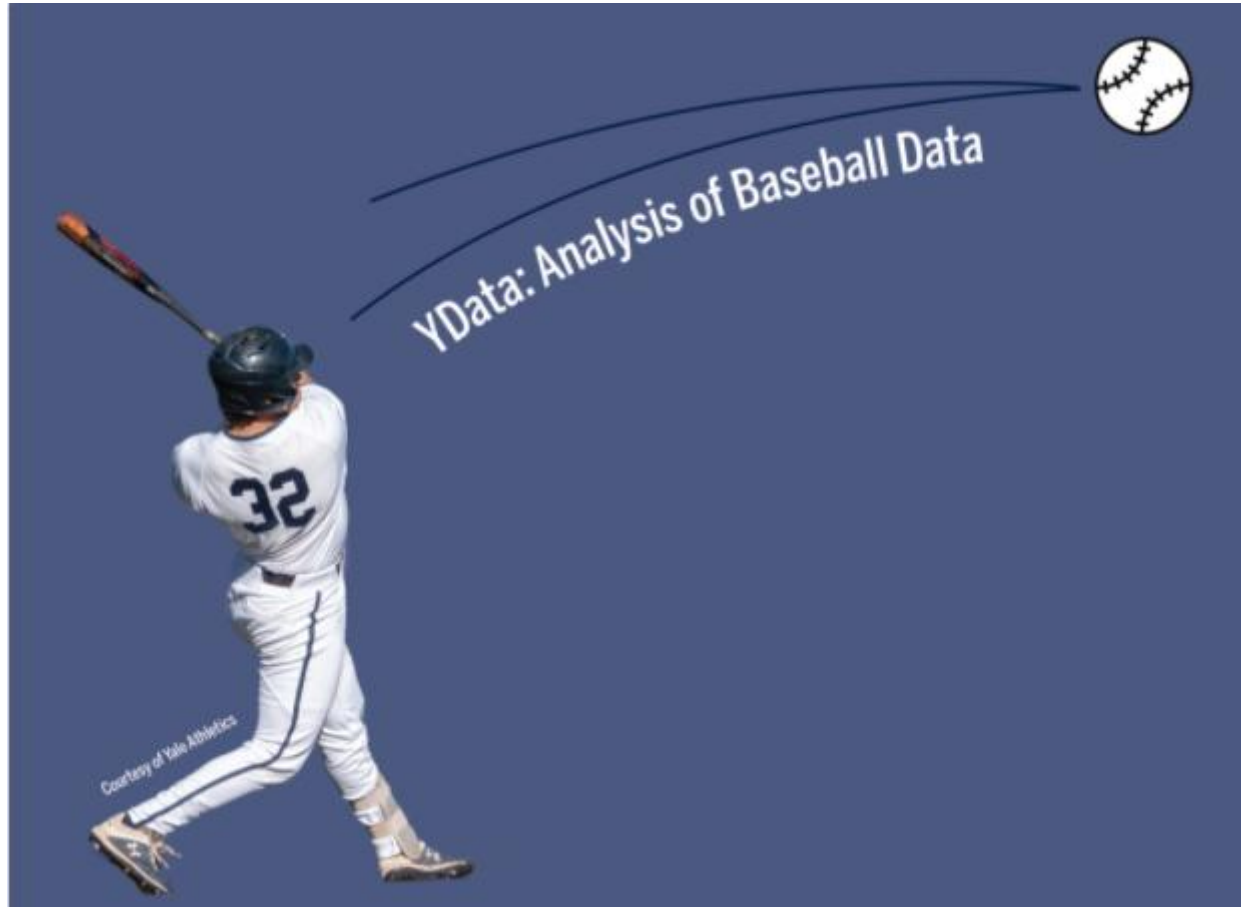


S&DS 1730



Class 10: Correlation and regression models

Overview

Discussion of chapter 7 of Astroball

Correlation

Simple linear regression

Multiple regression

Final projects!

Use the Data Science methods you have learned in this class to answer a question related to baseball!

You will give a 2 minute (one slide) presentation on your project during class on April 20th

Final project written reports are due at 11pm on April 26th

- Report should be a Jupyter notebook approximately 7 pages long
- A final project Jupyter notebook template has been uploaded to the class GitHub site



The baseball season has started!

AL EAST	W	L	PCT	GB
 Toronto Blue Jays	3	0	1.000	-
 New York Yankees	3	0	1.000	-
 Baltimore Orioles	2	1	.667	1.0
 Tampa Bay Rays	1	2	.333	2.0
 Boston Red Sox	1	2	.333	2.0

NL WEST	W	L	PCT	GB
 Los Angeles Dodgers	3	0	1.000	-
 San Diego Padres	1	2	.333	2.0
 Arizona Diamondbacks	0	3	.000	3.0
 Colorado Rockies	0	3	.000	3.0
 San Francisco Giants	0	3	.000	3.0

USA Today baseball predictions

MLB win totals for 2026

Our five-person panel's aggregate projections

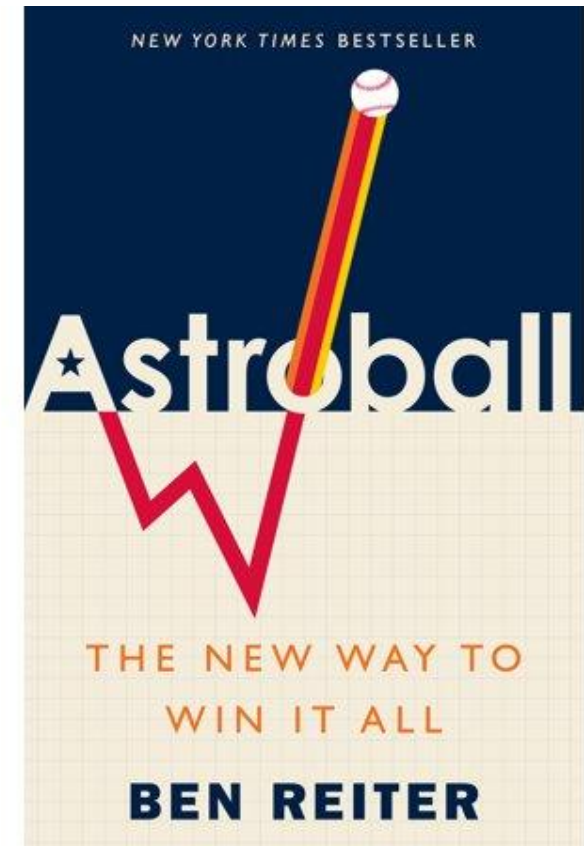
AMERICAN LEAGUE		
East	Central	West
 Blue Jays 91-71	 Tigers 90-72	 Mariners 92-70
 Yankees 90-72	 Guardians 83-79	 Rangers 86-76
 Orioles 85-77	 Royals 83-79	 Astros 84-78
 Red Sox 84-78	 White Sox 67-95	 Athletics 76-86
 Rays 76-86	 Twins 65-97	 Angels 72-90

NATIONAL LEAGUE		
East	Central	West
 Phillies 92-70	 Cubs 88-74	 Dodgers 98-64
 Mets 87-75	 Brewers 87-75	 Padres 85-77
 Braves 84-78	 Reds 83-79	 Giants 82-80
 Marlins 79-83	 Pirates 74-88	 Diamondbacks 80-82
 Nationals 62-100	 Cardinals 68-94	 Rockies 57-105

Discussion on chapter 7 of Astrobball

Let's discuss the chapter 7 for 3 minutes in small groups of 3-4 people and then have a larger conversation as a group

- E.g., Discuss your quote and reaction to chapter 7



Thoughts on the chapter 7 of Astrobball?

I couldn't find the paper "In Search of David Ross" presented at the Sloan Conference

I did find another paper by the same authors:

Journal of Sports Analytics 5 (2019) 247–279
DOI 10.3233/JSA-190248
IOS Press

Uncovering the sources of team synergy: Player complementarities in the production of wins

Scott A. Brave^{a,*}, R. Andrew Butters^b and Kevin A. Roberts^c

^aFederal Reserve Bank of Chicago, Economic Research, IL, Chicago

^bBusiness Economics and Public Policy, Kelley School of Business, Indiana University, IN, Bloomington

^cDepartment of Economics, Duke University, NC, Durham

$$\hat{\varepsilon}_{int} = \underbrace{w_{ii} f_{it} \lambda_n}_{\text{"Measurement Error"}} + \underbrace{\sum_{i:j \neq i} w_{ij} f_{jt} \lambda_n}_{\text{"Team Synergy"}}$$

$$\begin{aligned} \log(\text{salary}_{it}) &= \sum_c \gamma_c (FA_{cit} \sum_{n=1}^{t-1} WAR_{in}) + \sum_c \beta_c (FA_{cit} \sum_{n=1}^{t-1} pcWAR_{in}) + \\ &\sum_c \theta_c FA_{cit} + \sum_c \theta_c (FA_{cit} * teamExp_{it-1}) + \sum_p \rho_p pos_{pit} + \\ &\sum_p \phi_p (pos_{pit} * age_{it}) + \sum_p \lambda_p (pos_{pit} * mlbExp_{it-1}) + \\ &\sum_p \tau_p (pos_{pit} * mlbExp_{it-1}^2) + \alpha_i + \varepsilon_{it}, \end{aligned} \quad (17)$$

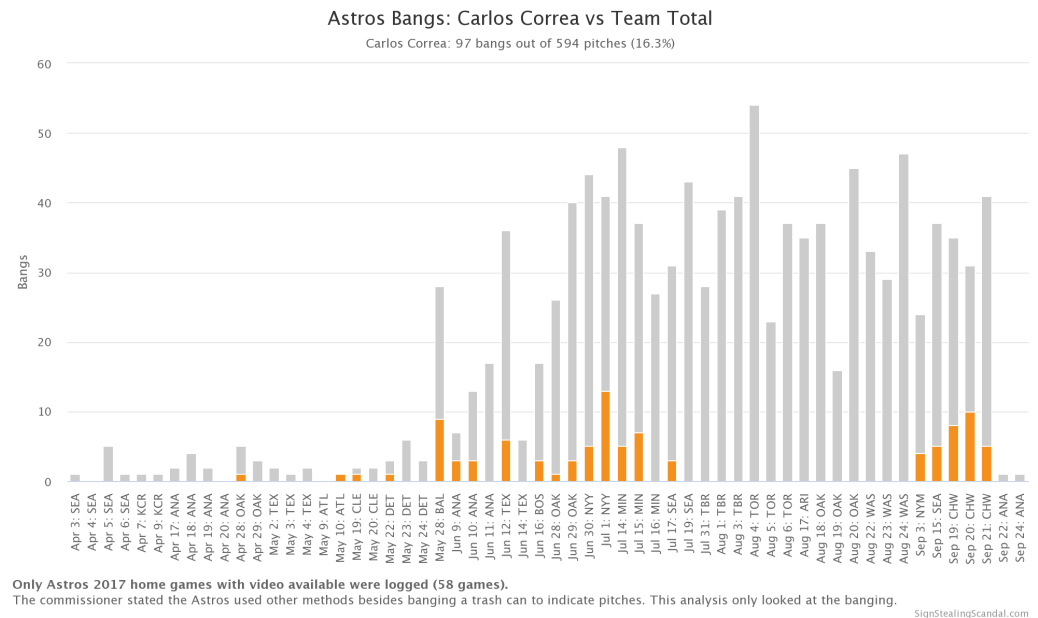
Thoughts on the chapter 7 of Astrobball?

“Beltran’s impact was impossible for the Nerd Cave to quantify, but Carlos Correa attempted to attach a number to it: seven.

Of the 24 home runs he hit in 2017, by the end of the regular season Correa attributed precisely seven to Beltran’s influence, and to Beltran’s showing him how to use video to break down opposing pitchers to a depth Correa had never before imagined, to his identifying their tells.”

pg 160

Are these home runs related to trashcan bangs and can we find these home runs?



Thoughts on the chapter 7 of Astrobball?

Burial of Beltran's glove



ESPN: Is any team having more fun than the Houston Astros?

Relationships between two quantitative variables

Do power hitters strike out more often?

Chris Davis in 2013:



53 home runs



199 strike outs

Scatter plots

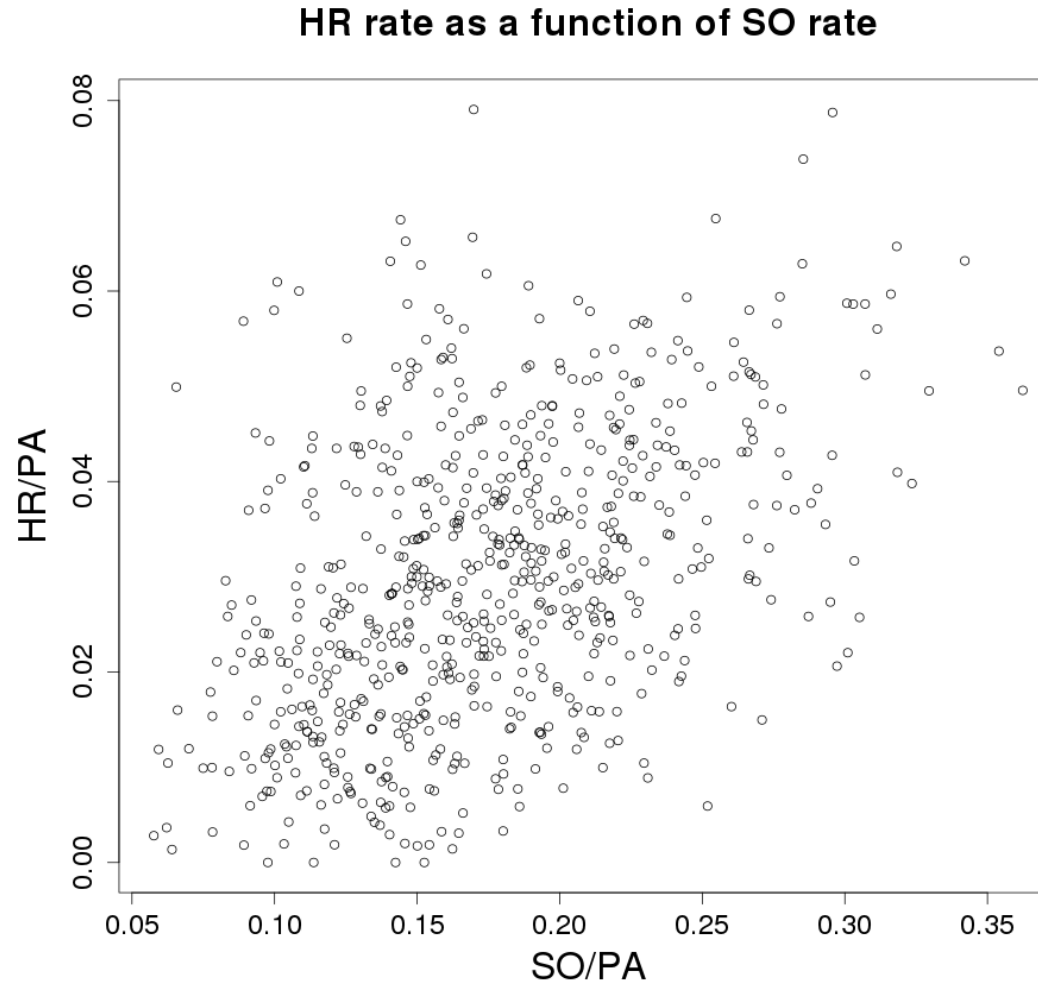
A **scatterplot** graphs the relationship between two variables

Each axis represents the value of one variables.

Each point the plot shows the value for the two variables for a single data case.

If there is an explanatory and response variable, then the explanatory variable is put on the x-axis and the response variable is put on the y-axis

Scatter plots: 2010-2014 home run rate as a function of strike out rate



```
plt.scatter(x_sequence, y_sequence)
```

Questions when looking at scatter plots

Do the points show a clear trend

Does it go upward or downward?

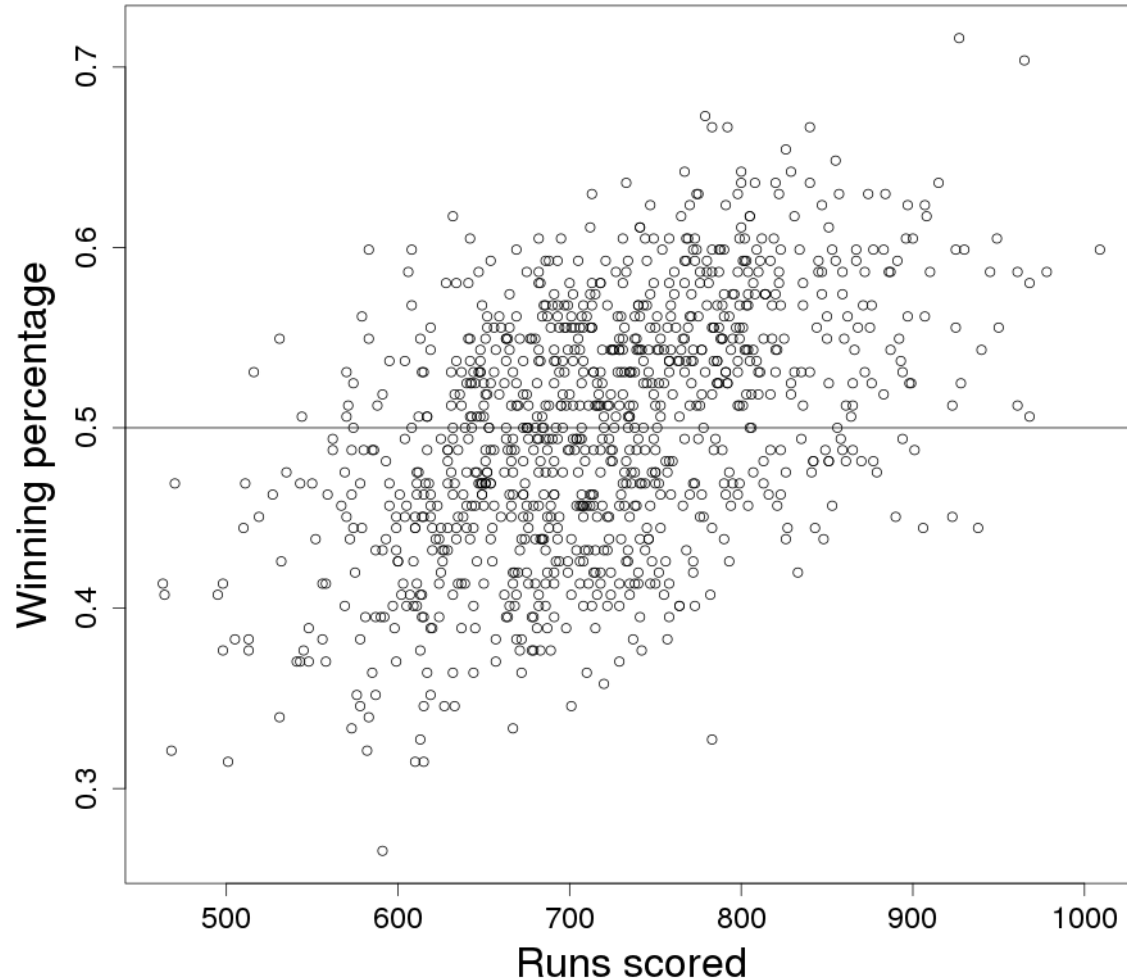
How much scatter around the trend?

Does the trend seem be linear (follow a line) or is it curved?

Are there any outlier points?

Relationship between runs scored and wins

(each point is one team from one season)



Do the points show a clear trend

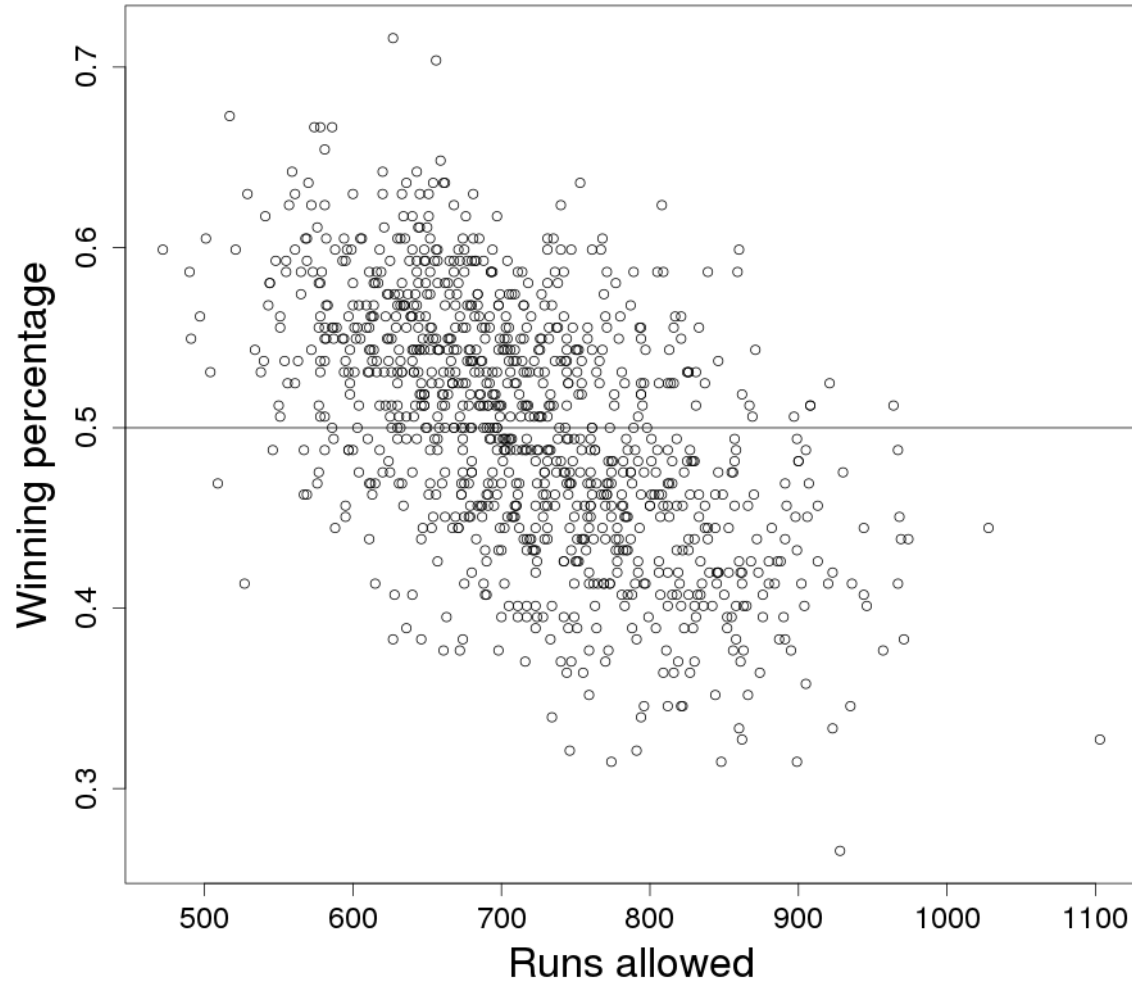
Does it go upward or downward?

How much scatter around the trend?

Does the trend seem be linear (follow a line) or is it curved?

Are there any outlier points?

Relationship between runs allowed and wins



Do the points show a clear trend

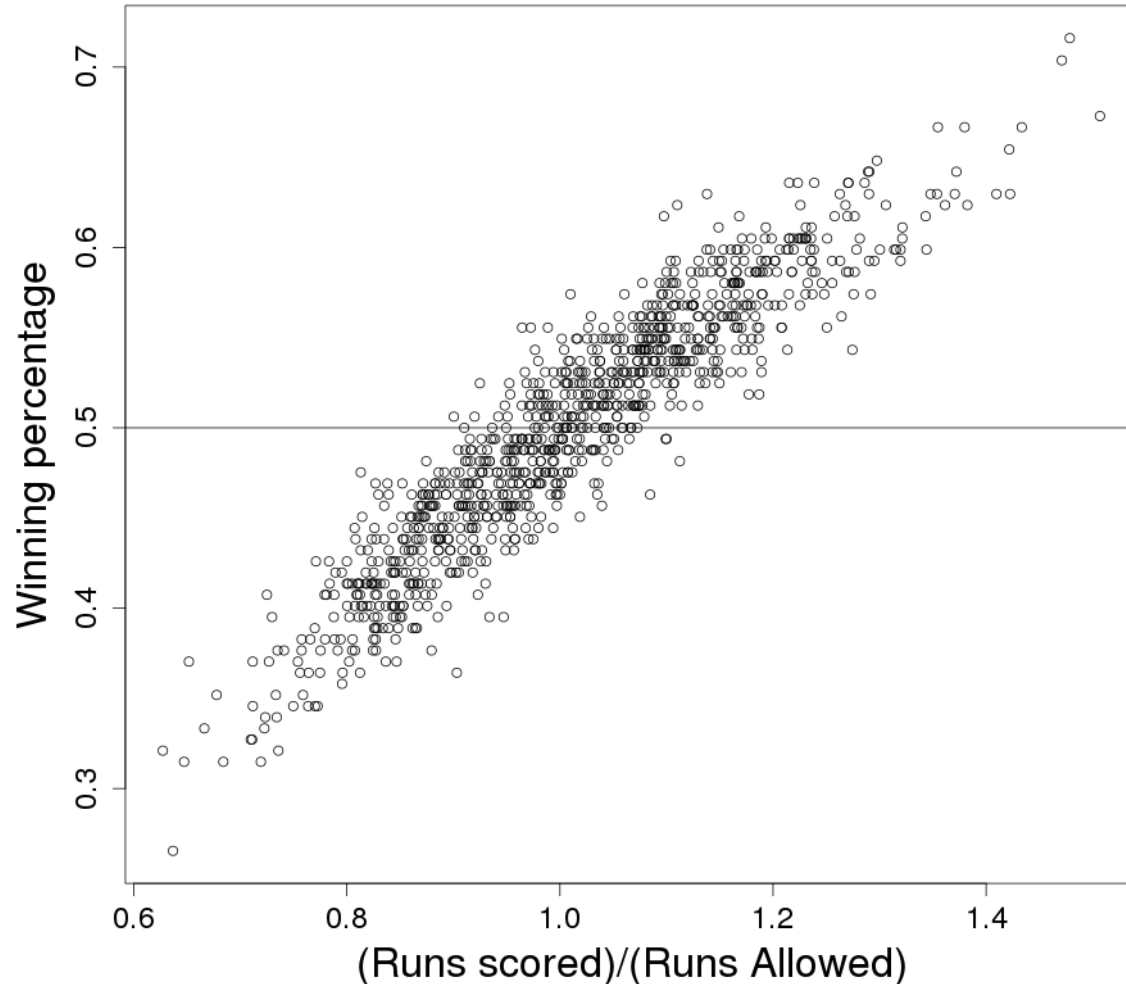
Does it go upward or downward?

How much scatter around the trend?

Does the trend seem be linear (follow a line) or is it curved?

Are there any outlier points?

Relationship between (runs scored)/(runs allowed) and wins



Do the points show a clear trend

Does it go upward or downward?

How much scatter around the trend?

Does the trend seem be linear (follow a line) or is it curved?

Are there any outlier points?

Correlation

The **correlation** is measure of the strength and direction of a linear association between two variables.

$$r = \frac{1}{(n-1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

```
my_series.corr(my_second_series)
```

```
np.corrcoef(sequence1, sequence2)
```

Properties of the correlation

Correlation is always between -1 and 1: $-1 \leq r \leq 1$

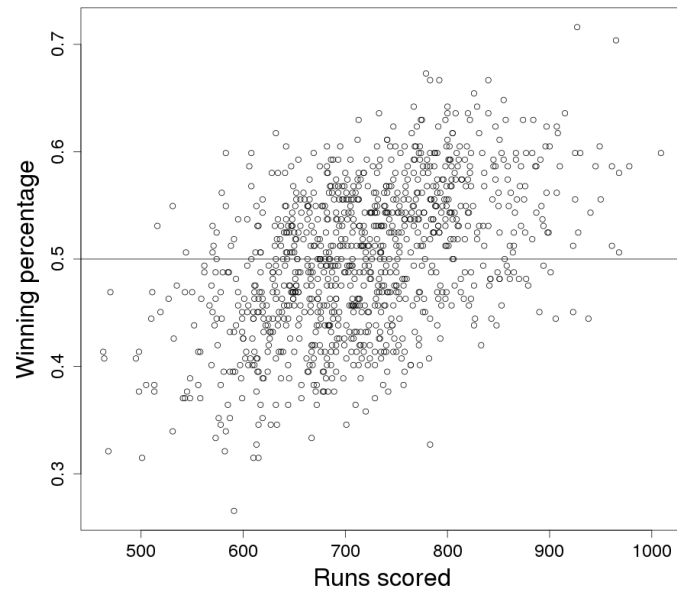
The sign of r indicates the direction of the association

Values close to ± 1 show strong linear relationships,
values close to 0 show no linear relationship

Correlation is symmetric: $r = \text{cor}(x, y) = \text{cor}(y, x)$

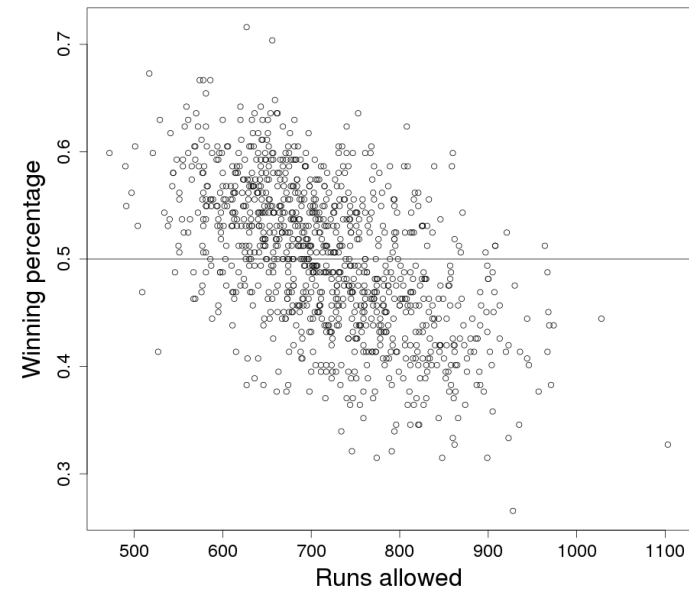
Correlation between runs and wins

Runs scored and wins



$r = .50$

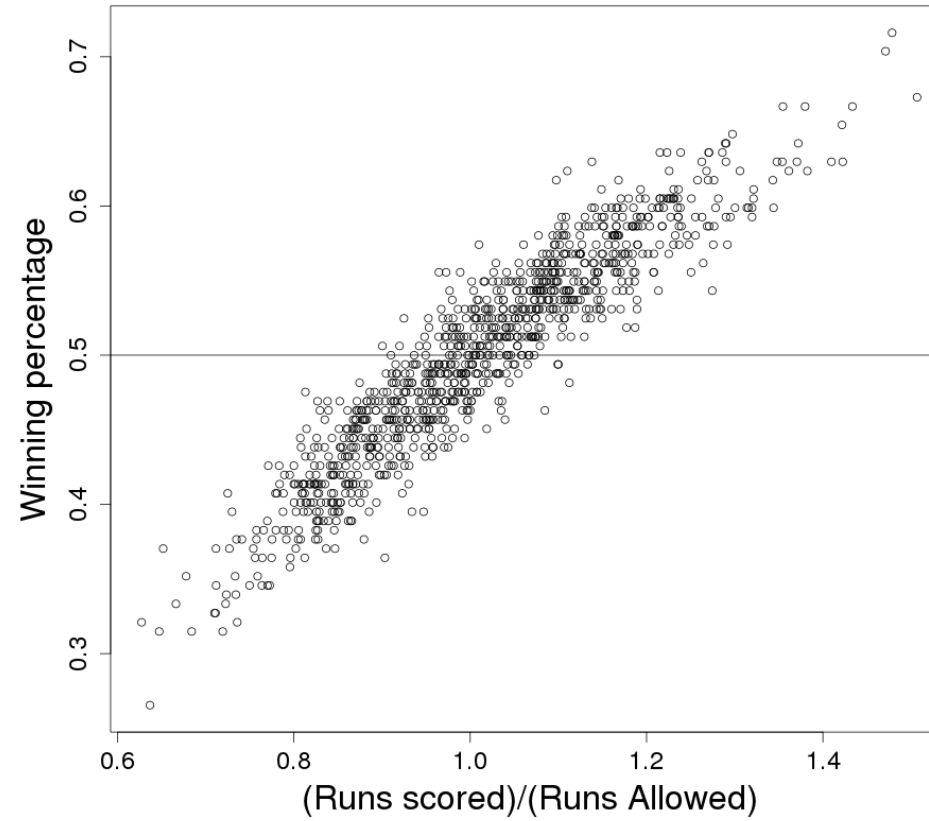
Runs allowed and wins



$r = -.55$

Correlation between runs and wins

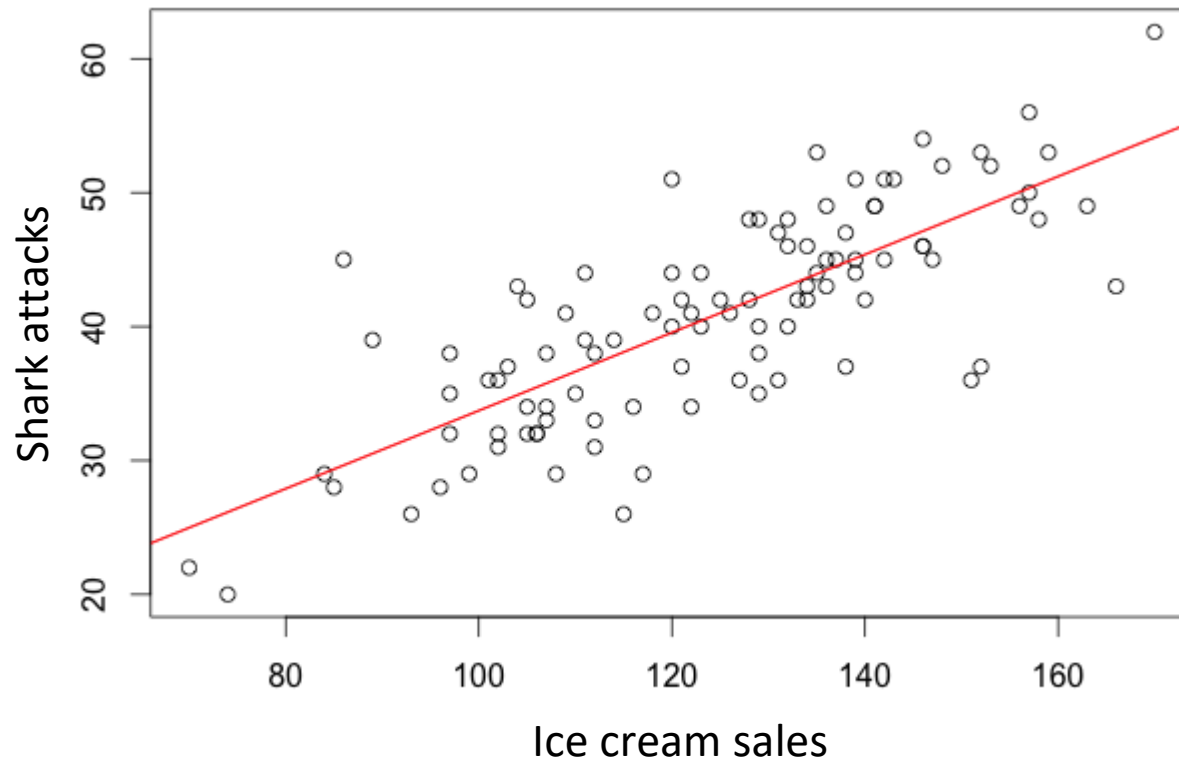
(runs scored)/(runs allowed) and wins



$r = .93$

Correlation caution #1

A strong positive or negative correlation does not (necessarily) imply a **cause and effect** relationship between two variables



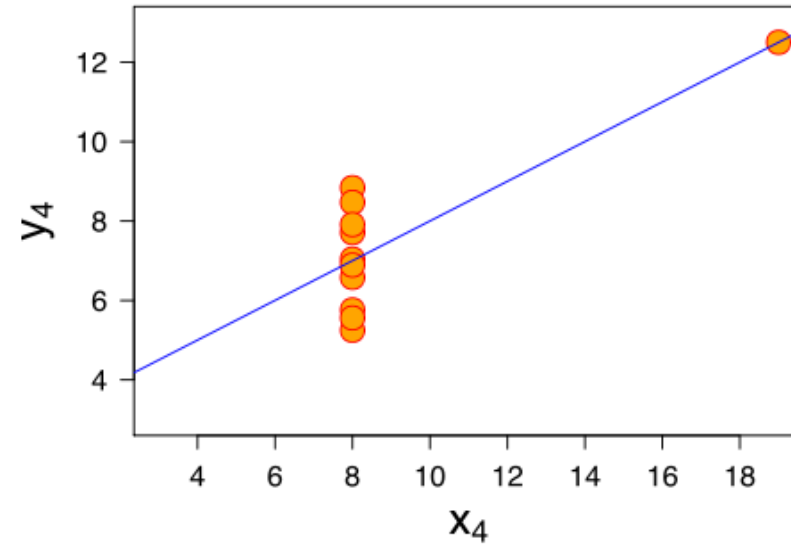
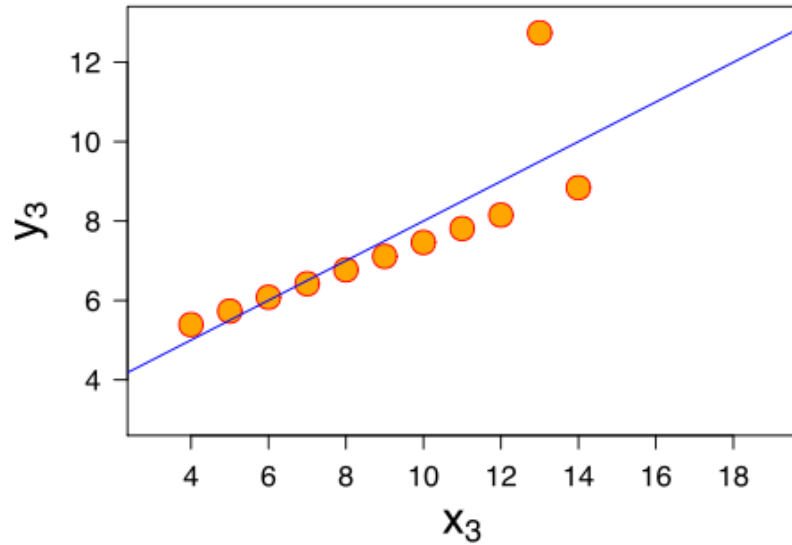
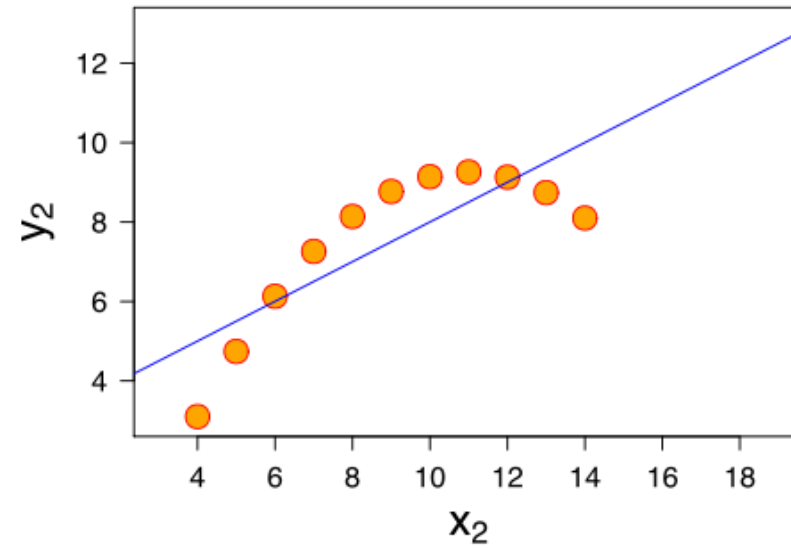
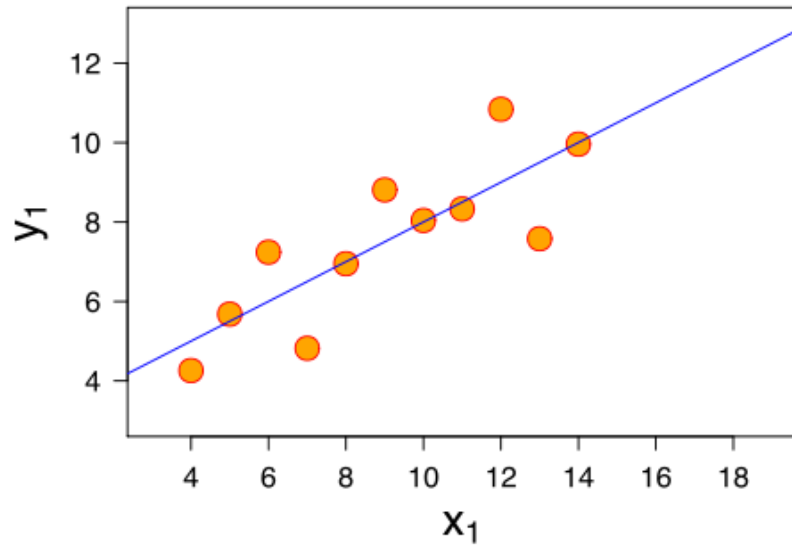
Correlation caution #2

A correlation near zero does not (necessarily) mean that two variables are not associated. Correlation only measures the strength of a linear relationship

Correlation caution #3

Correlation can be heavily influenced by outliers. **Always plot your data!**

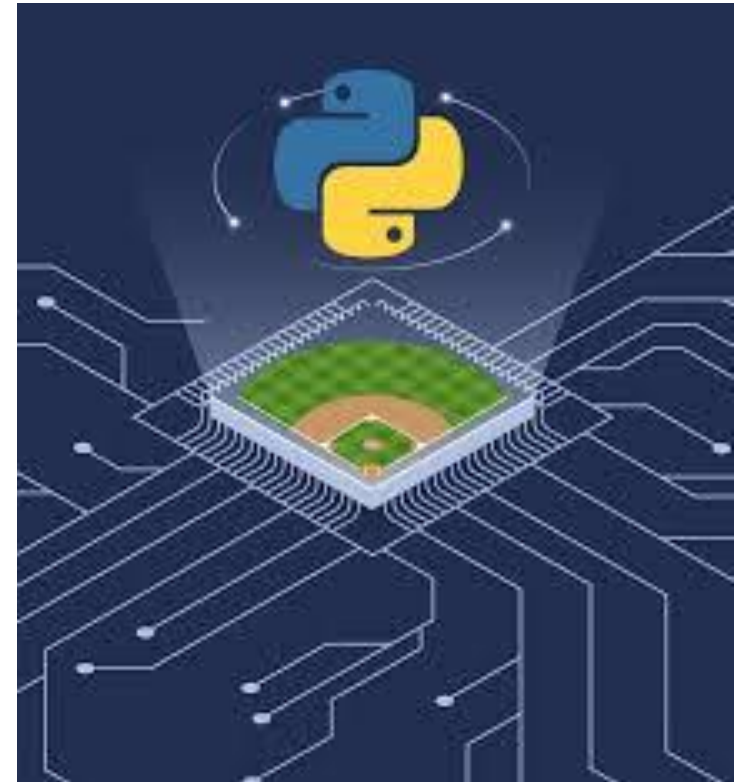
Anscombe's quartet ($r = 0.81$)



Scatter plots and correlation in Python

Let's explore scatter plots and correlation in Python by looking at the relationship between:

1. The number of wins a team has in a season
2. The number of runs the team scored



Who is a better hitter: Derek Jeter or David Ortiz?



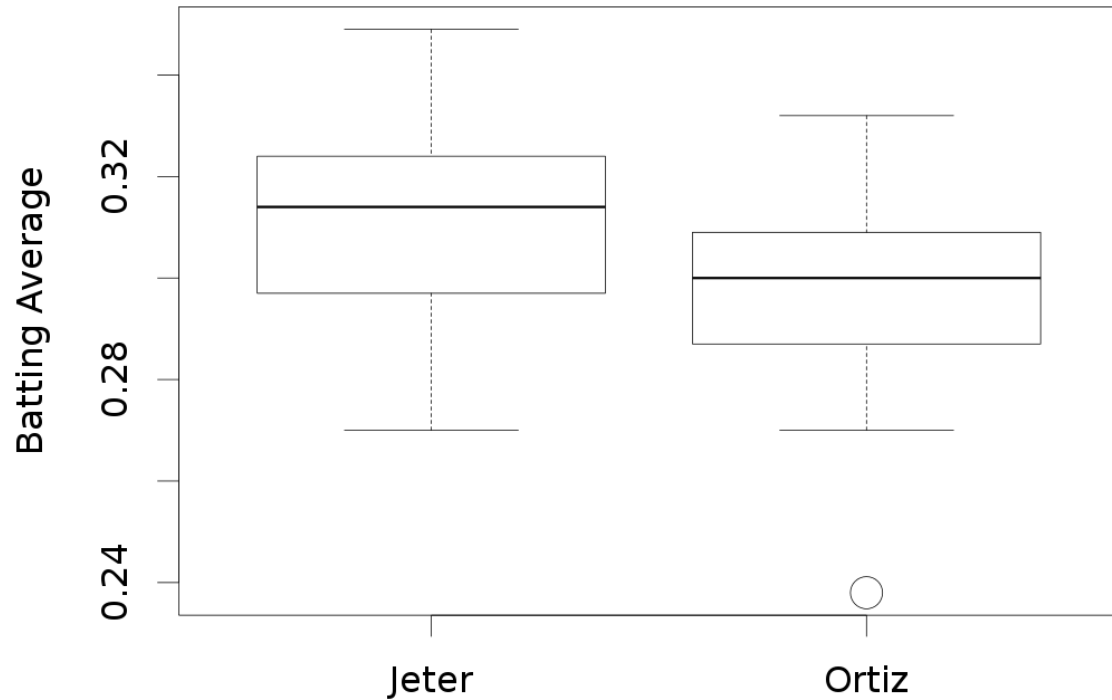
Derek Jeter



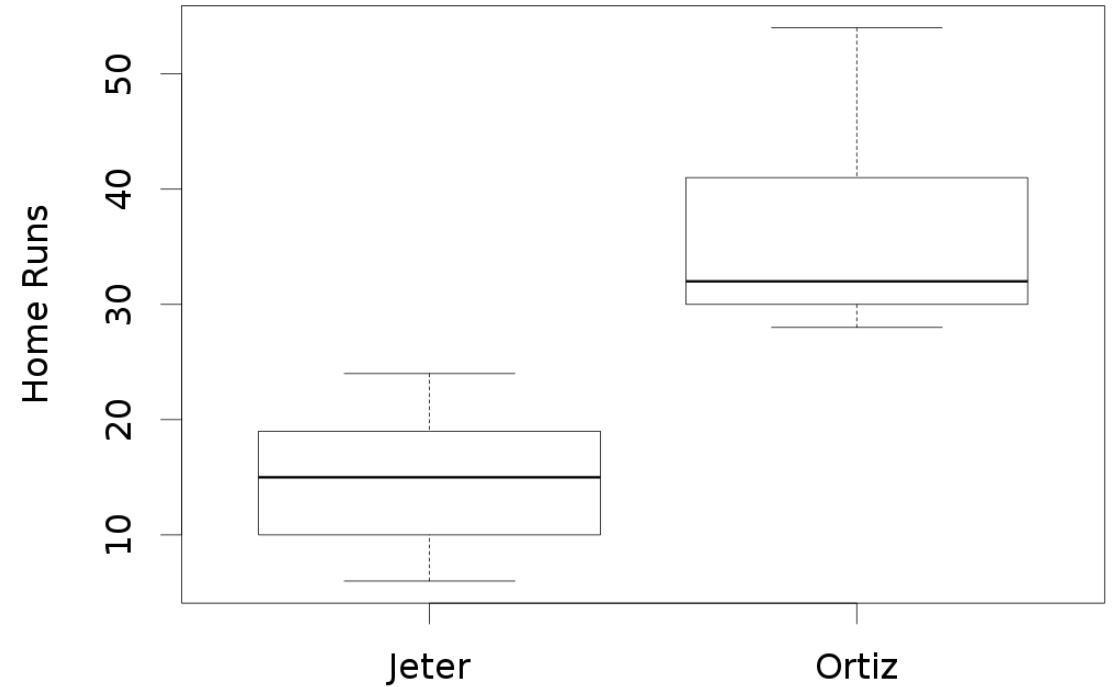
[David Ortiz](#)

How can I prove to my misguided Yankee fan friends that Ortiz is better?

Who is a better hitter: Derek Jeter or David Ortiz?



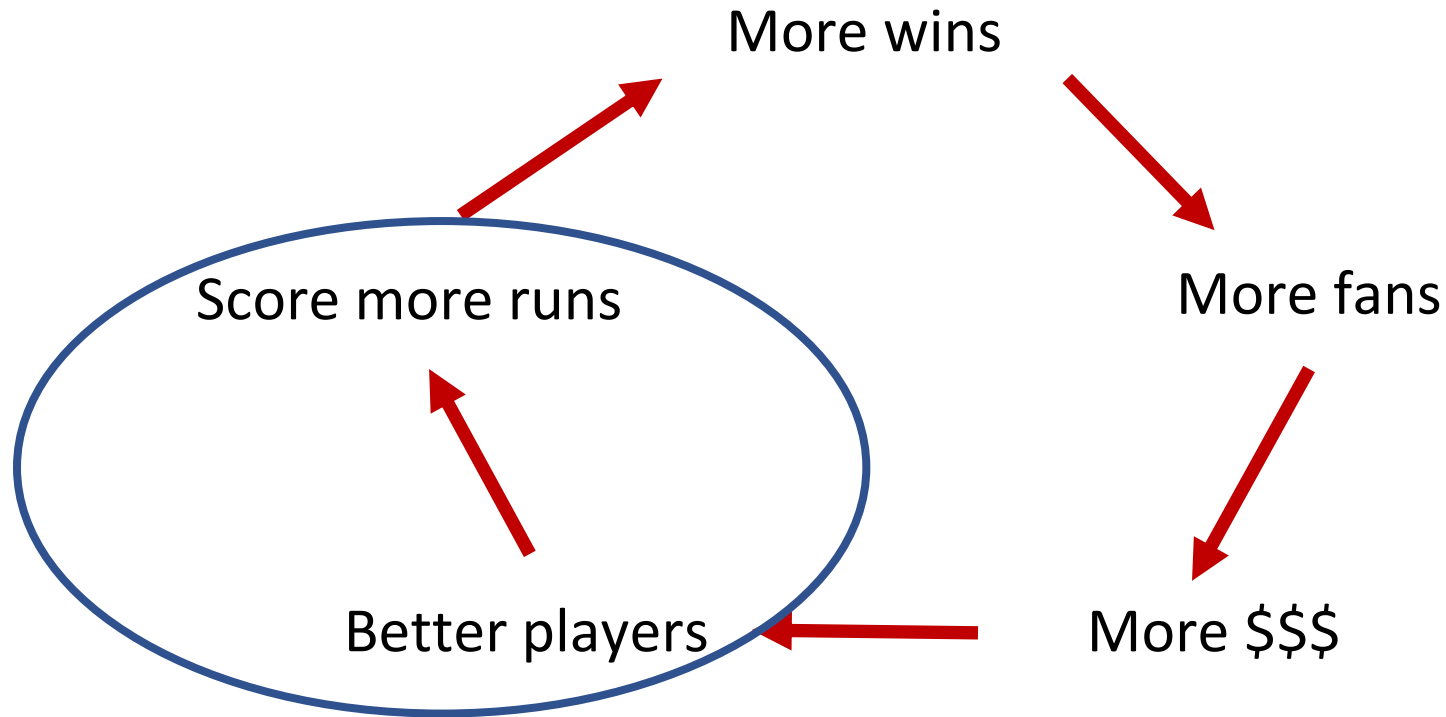
Jeter has a better batting average



Ortiz hits more home runs

Is power or batting average more important?

The great cycle of baseball



We can evaluate how 'good' a statistic is based on how well it correlates with the number of runs a team scores

What is the best statistic to use?

One idea: the 'best' statistic to judge a player is the statistic that is most correlated with runs

- We can then use this to examine how good a hitter is

Descriptive statistics we can evaluate by correlating with runs:

HR: Home runs

BA: Batting Average: H/AB

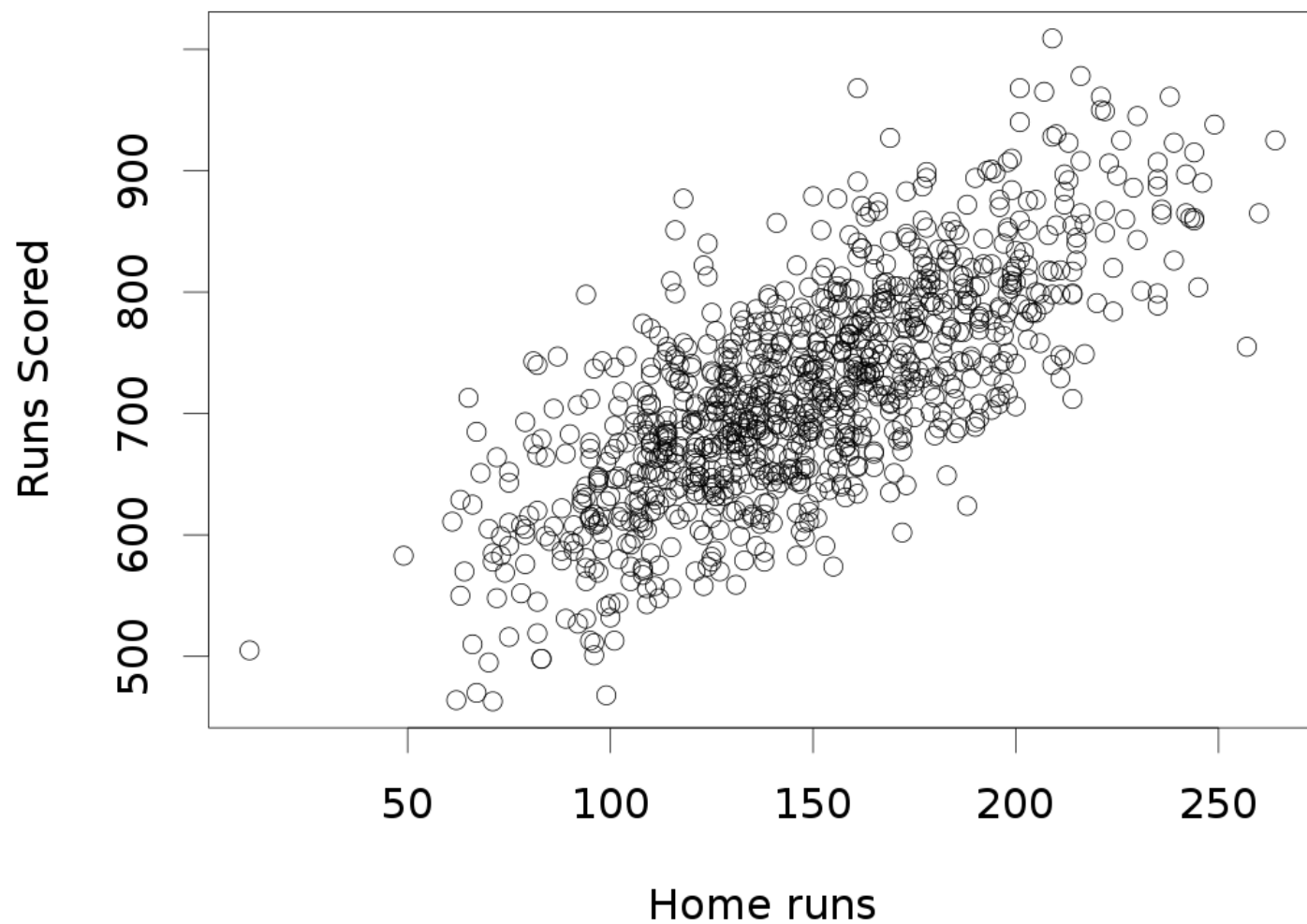
OBP: On-Base Percentage: $(H + BB)/PA$

SLG: Slugging percentage: $(1 \cdot 1B + 2 \cdot 2B + 3 \cdot 3B + 4 \cdot HR)/AB$

On-base plus Slugging (OPS): $OBP + SLG$

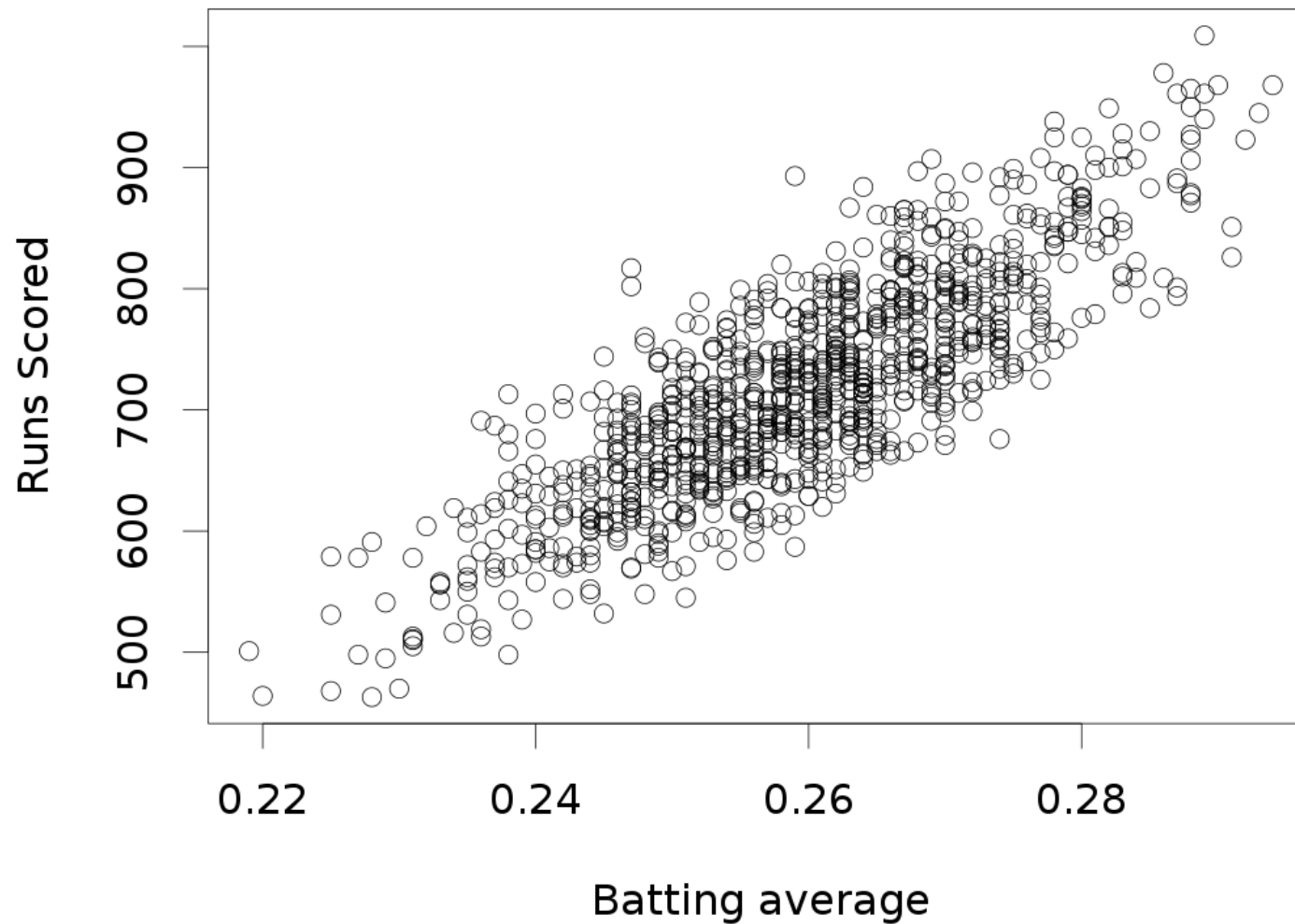
Correlation between HR and runs

$r = 0.74$



Correlation between BA and runs

$r = 0.83$



Lab 8

On lab 8 you will calculate the correlation between the number of runs a team scored and team level statistics for HR, BA, SLG, OBP, and OPS

Whichever statistic has the highest correlation with runs we will say is the best statistics for capturing batting ability

We can then compare Ortiz and Jeter (and potentially other players) using this statistic to determine who is the better hitter



[Onion infographic](#)

[Other Onion articles](#)



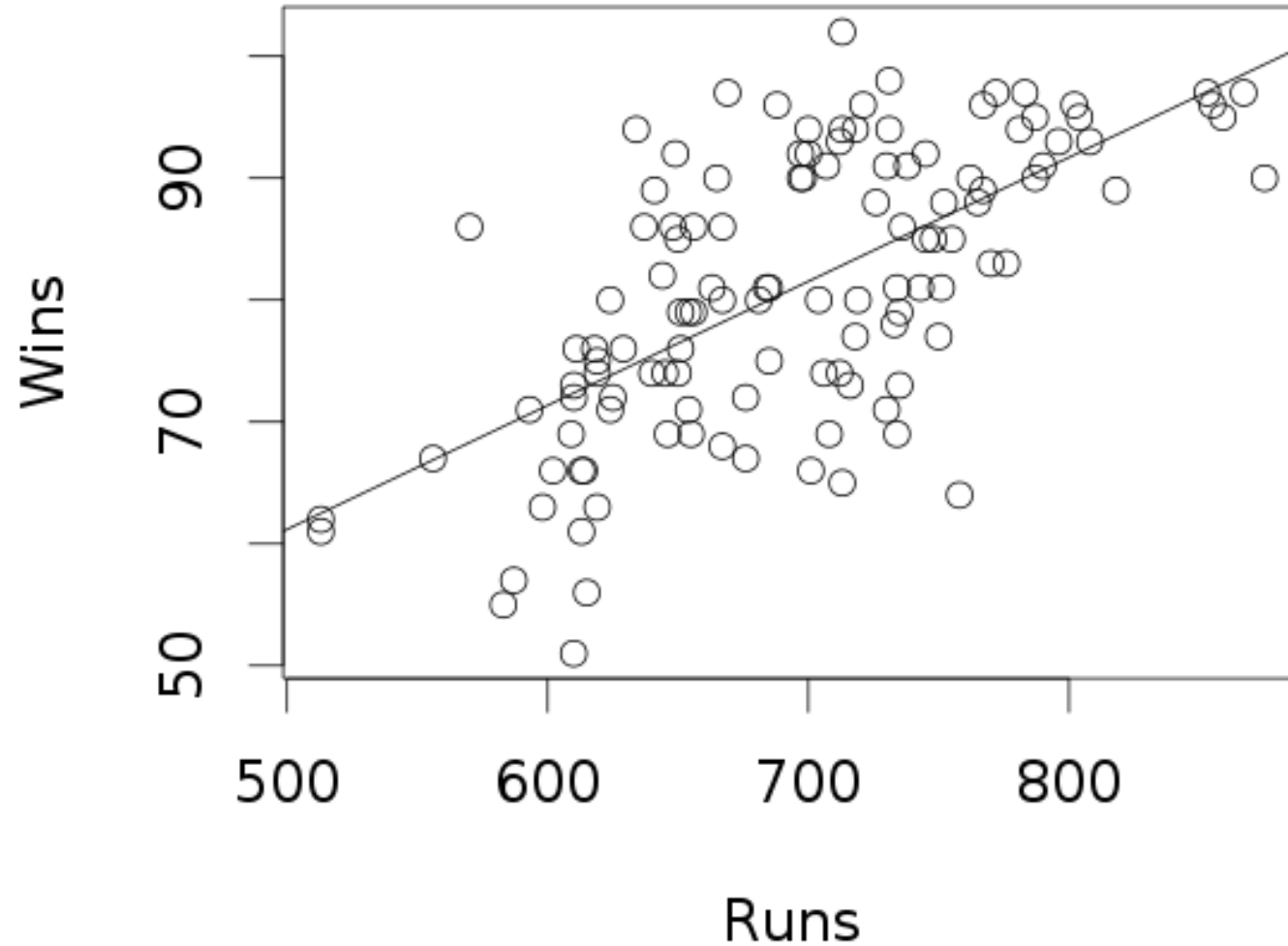
Linear regression

Regression

Regression is method of using one variable to predict the value of a second variable

In **linear regression** we fit a line to the data to make predictions, called the **regression line**

Regression line: wins as a function of runs scored



Equation for a line

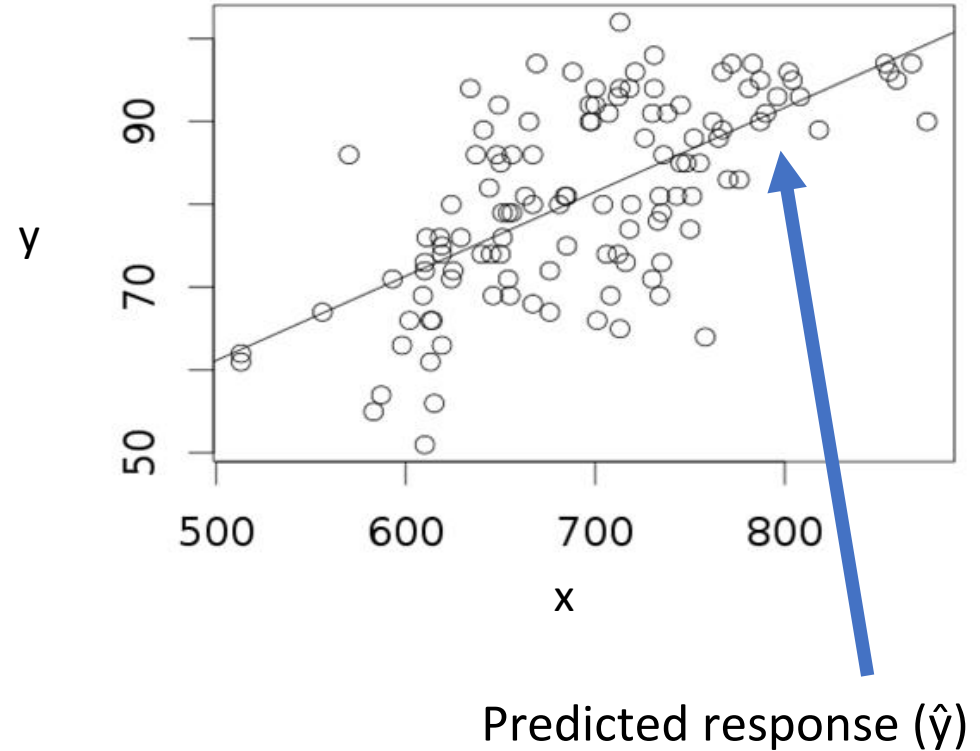
$$\hat{y} = b_0 + b_1 \cdot x$$

$$\hat{y} = f(x)$$

coefficients
(constants)

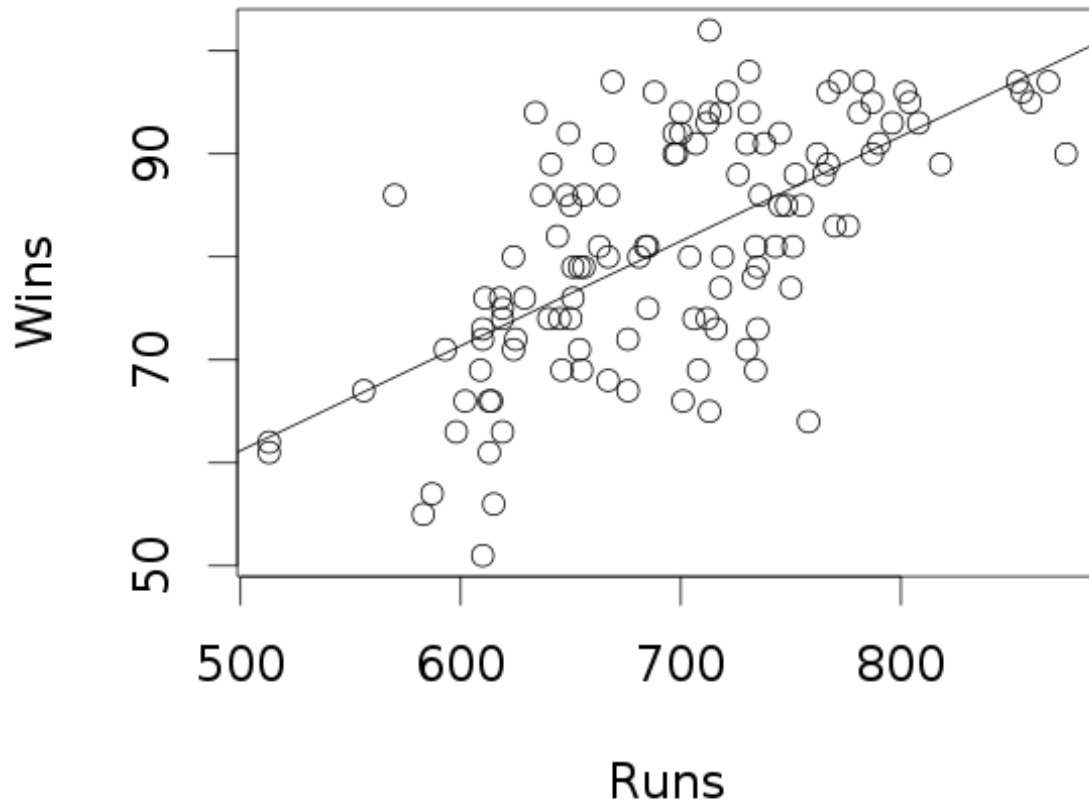
Predicted response ($\hat{y}$)

Explanatory variable (x)



$\hat{y}$ is our prediction (best guess) of what we think y should be based on the value x

Wins runs regression



$$\hat{y} = b_0 + b_1 \cdot x$$

$$b_0 = 14.47$$

$$b_1 = .088$$

$$\hat{w} = 14.47 + .088 \cdot runs$$

Q: If a team scores 700 runs, how many wins ($\hat{w}$) would we predict?

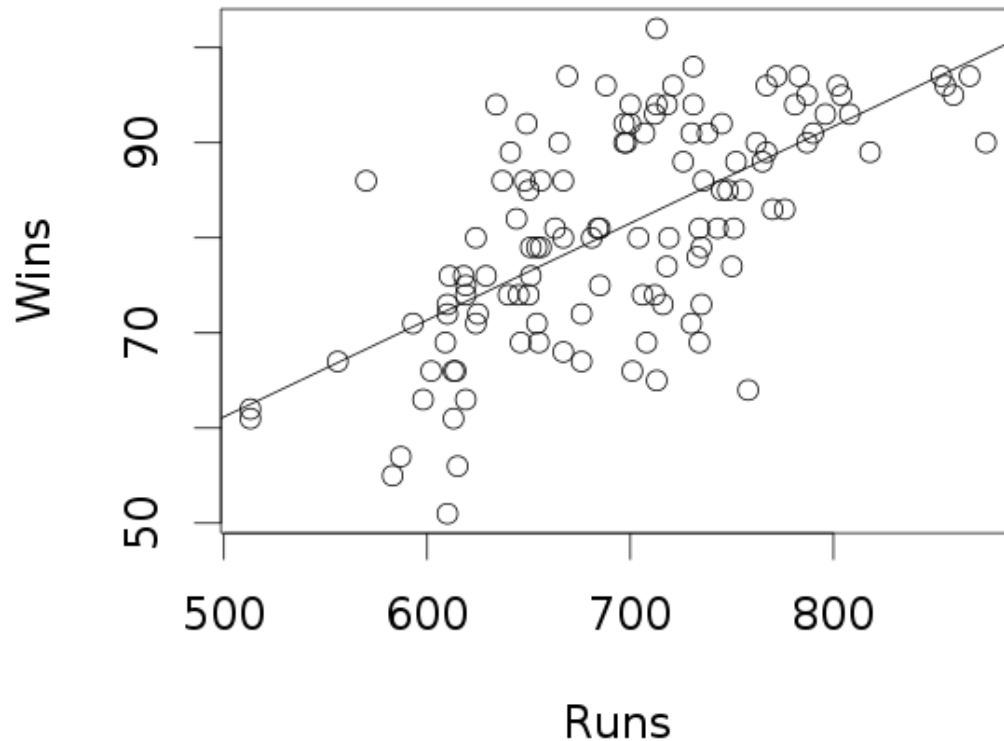
Interpreting the slope and intercept

$$\hat{y} = b_0 + b_1 \cdot x$$

The slope b_1 represents the predicted change in the response variable y given a one unit change in the explanatory variable x

The intercept b_0 represented the predicted value of the response variable y if the explanatory variable x were 0

Using the regression line to make predictions



$$\hat{y} = b_0 + b_1 \cdot x$$

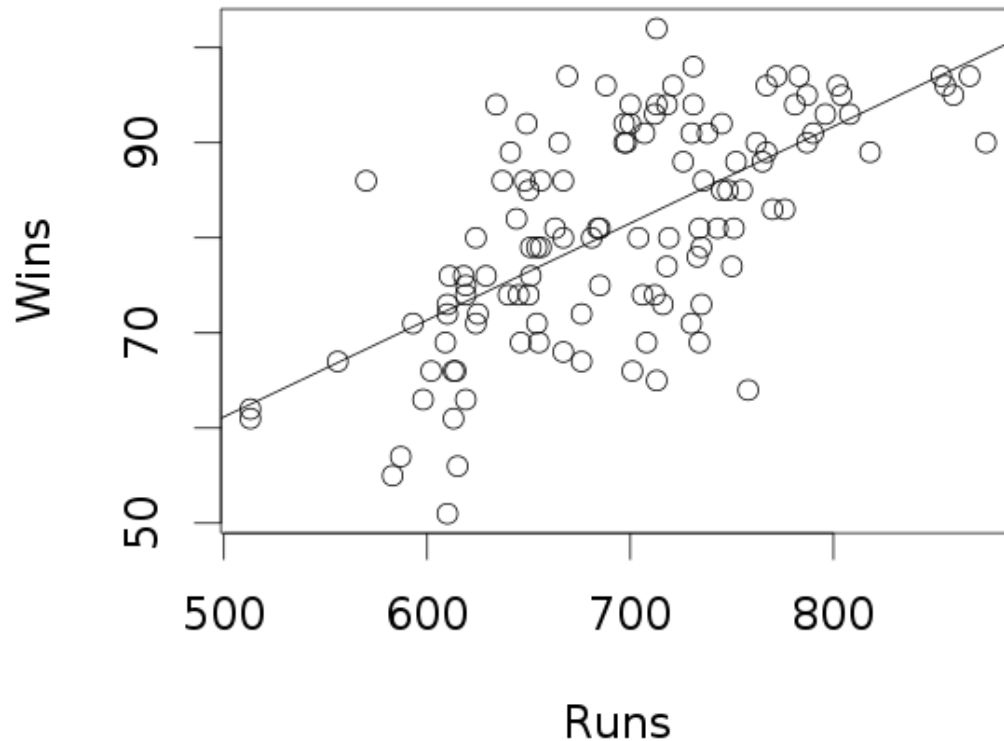
$$b_0 = 14.47$$

$$b_1 = .088$$

$$\hat{w} = 14.47 + .088 \cdot \text{Runs}$$

1. Approximately how many additional runs do you need to score for an additional win?
2. How many wins will you have if you score 0 runs all season?

Using the regression line to make predictions



$$\hat{y} = b_0 + b_1 \cdot x$$

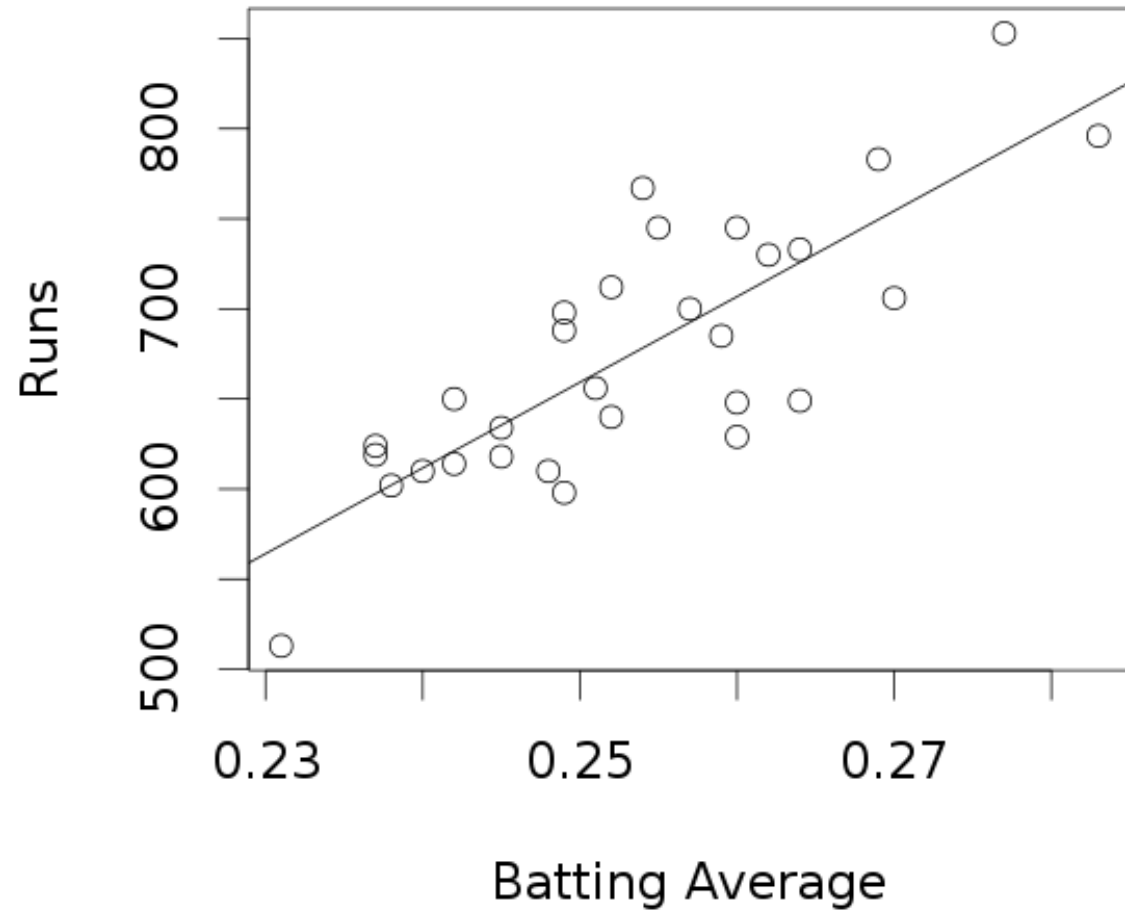
$$b_0 = 14.47$$

$$b_1 = .088$$

$$\hat{w} = 14.47 + .088 \cdot \text{Runs}$$

1. An additional win for ~11 additional runs scored
2. There will be 14.47 wins if you score 0 runs all season

Example 2: Runs and BA regression



$$\hat{y} = b_0 + b_1 \cdot x$$

$$b_0 = -770.2$$

$$b_1 = 5739.5$$

Write the equation for predicting runs as a function of BA

$$\hat{r} = -770.2 + 5739.5 \cdot BA$$

If a team had a batting average of 0.270, how many runs would you expect in a season?

$$\hat{r} = -770.2 + 5739.5 \cdot BA$$

$$\hat{y} = b_0 + b_1 \cdot x$$

$$\hat{r} = -770.2 + 5739.5 \cdot .270$$

$$b_0 = -770.2$$

$$\hat{r} = 772.465$$

$$b_1 = 5739.5$$

How about if a team batting .250?

Finding the coefficients b_0 and b_1

There are there a few popular Python libraries that are used for creating linear regression models which are:

- Numpy: `b1, b0 = np.polyfit(ndarray_x, ndarray_y, degree)`
- Scikit-learn
- Statsmodels

For now we will use the “formula notation” from Statsmodels package:

```
import statsmodels.formula.api as smf
```

Linear regression in Python

We can build a linear model using the Statsmodels syntax:

```
lm = smf.ols('y ~ x', data = my_df).fit()
```

We can extract $\mathbf{b}_0$ (the intercept) and $\mathbf{b}_1$ (the slope) from the model using:

```
coefficients = lm.params
```

Linear regression in Python

Building the linear model:

```
lm = smf.ols('R ~ BA', data = teams_162).fit()
```

Get the coefficients:

```
the_params = lm.params
```

```
Intercept    -526.921684  
BA            4744.561329  
dtype: float64
```

Can you write an equation using these coefficients?

$$A: \quad \hat{r} = -526.9 + 4744.5 \cdot BA$$

Making predictions

Once we have fit the regression model (and have the regression equation) we can use it to make predictions

To do this we can use the use:

```
sm_predictions = lm.predict(the_data)
```

Where `the_data` is a DataFrame that has as column with the same name as the x variable that was used when the model was fit

BA
.252
.263
.245
.255

Making predictions

We can also write our own function to make predictions based on the regression coefficients that were learned

```
def make_predictions(the_coefs, x_vals):
```

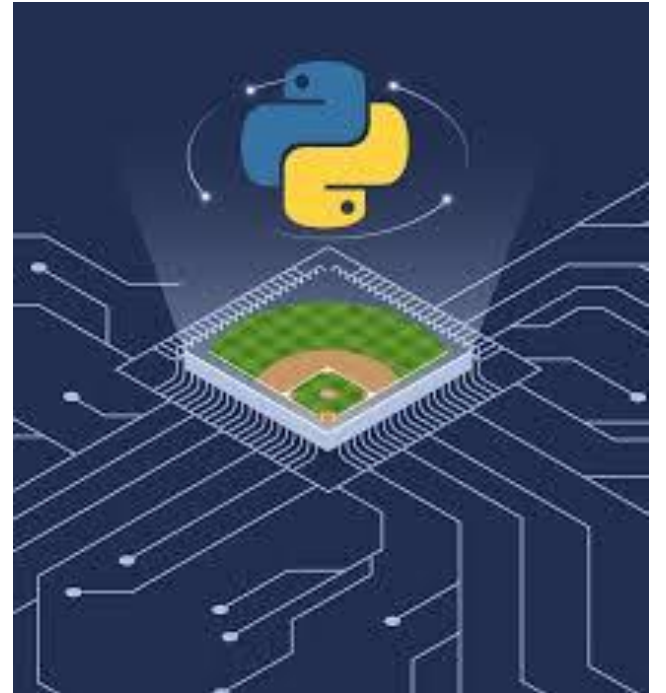
```
    ...
```

You will do this on part 2 of the lab!

Simple linear regression models in Python

Let's fit a linear regression model that predicts:

- The number of wins a team has in a season (y)
- Based on the number of runs the team scored (x)



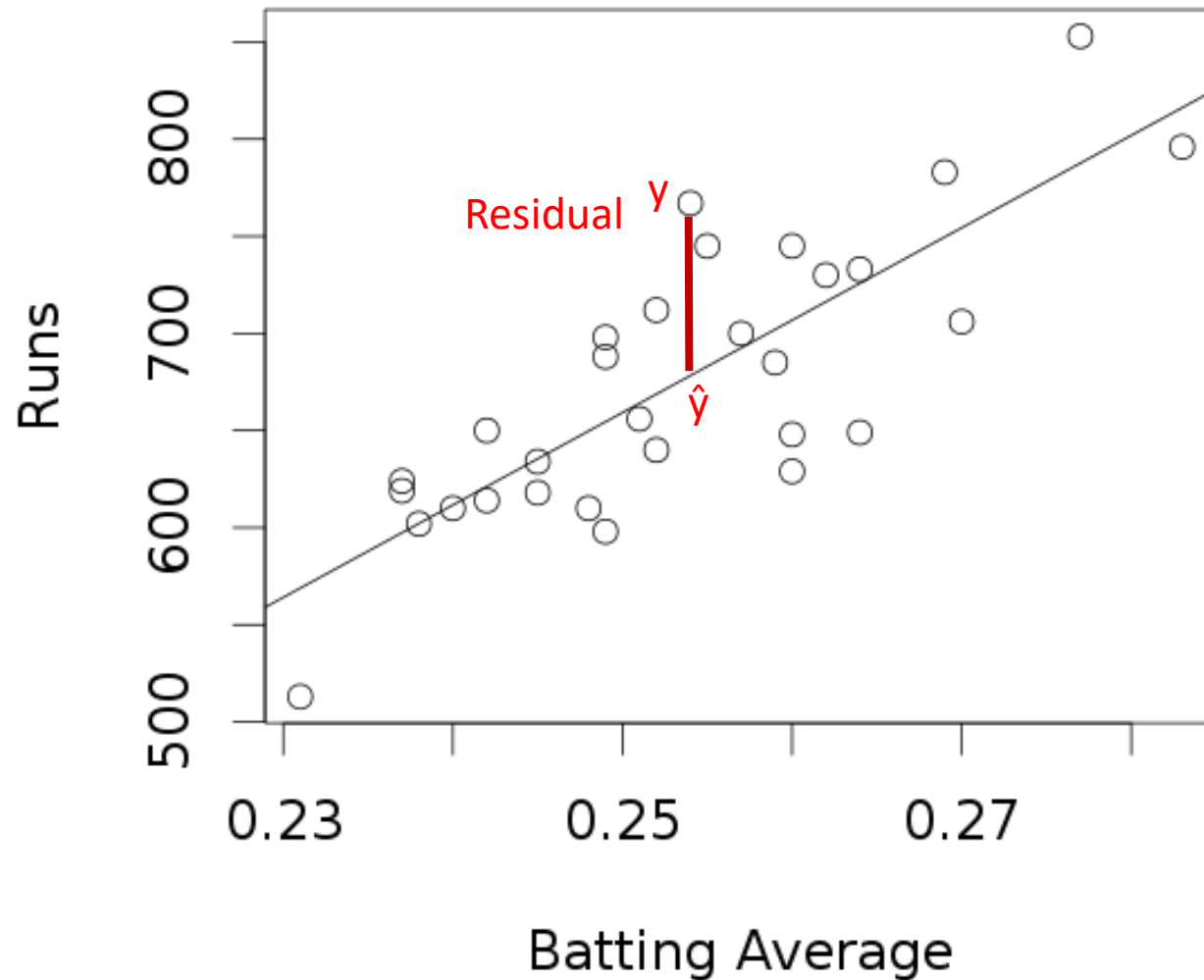
Residuals and the root mean squared error

Residuals

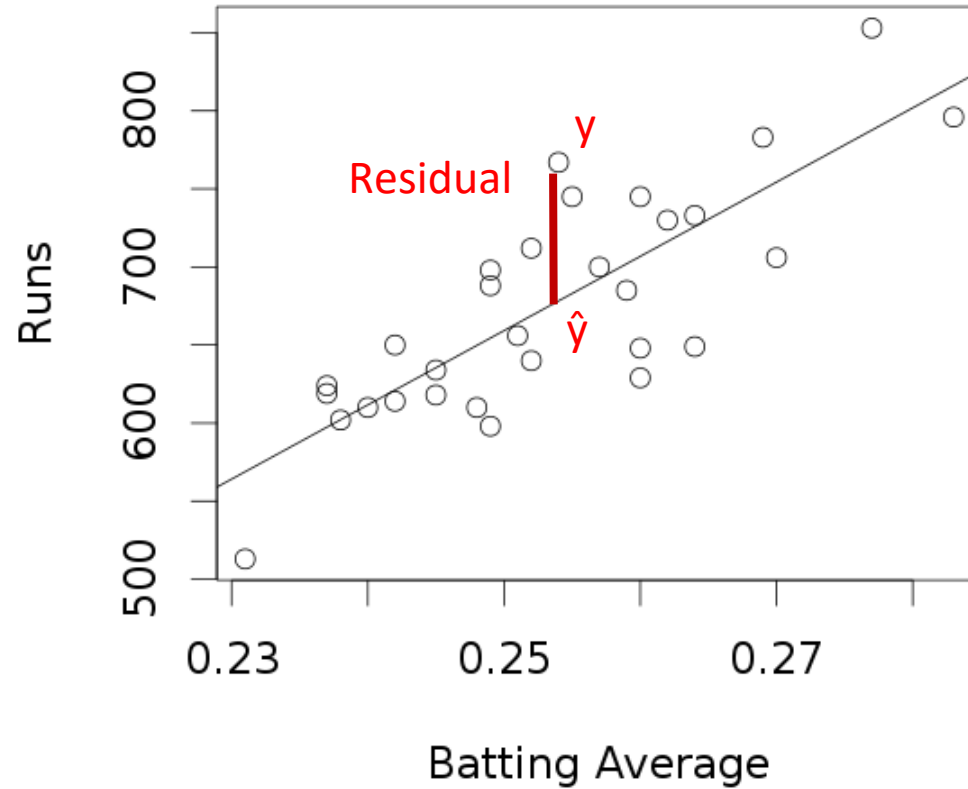
The **residual** at a data value is the difference between the observed (y) and predicted value ($\hat{y}$) of the response variable

$$\text{Residual} = \text{Observed} - \text{Predicted} = y - \hat{y}$$

Run vs. batting average (2013)

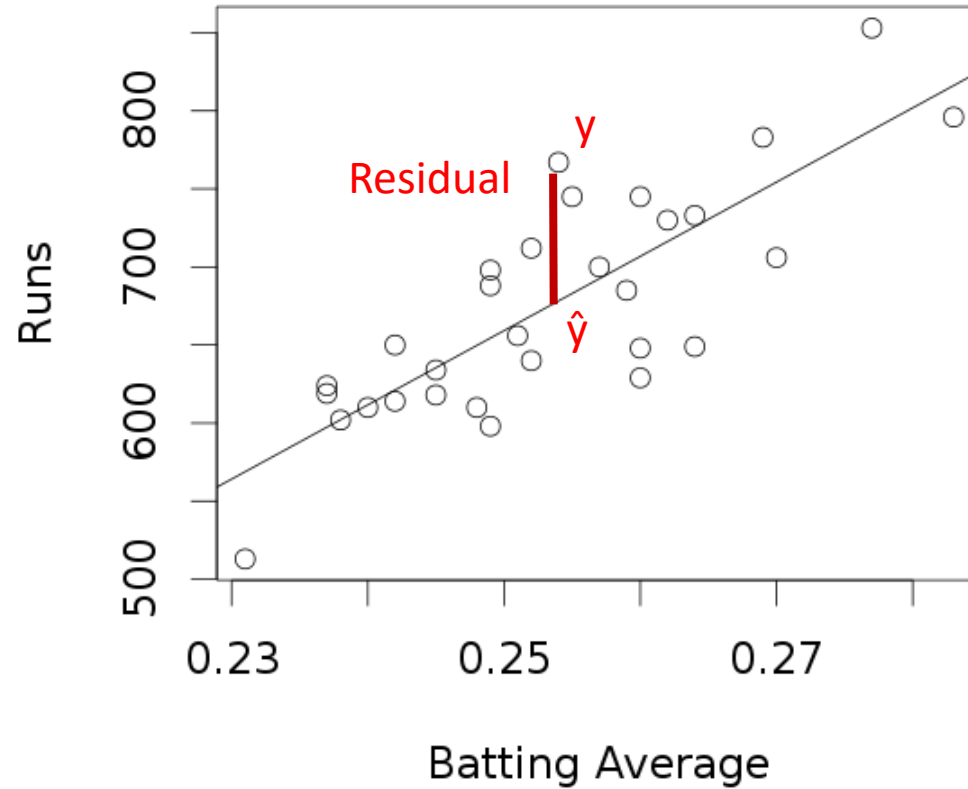


Measuring goodness of fit



If the residuals are small, then the line does a good job describing the data

Measuring goodness of fit

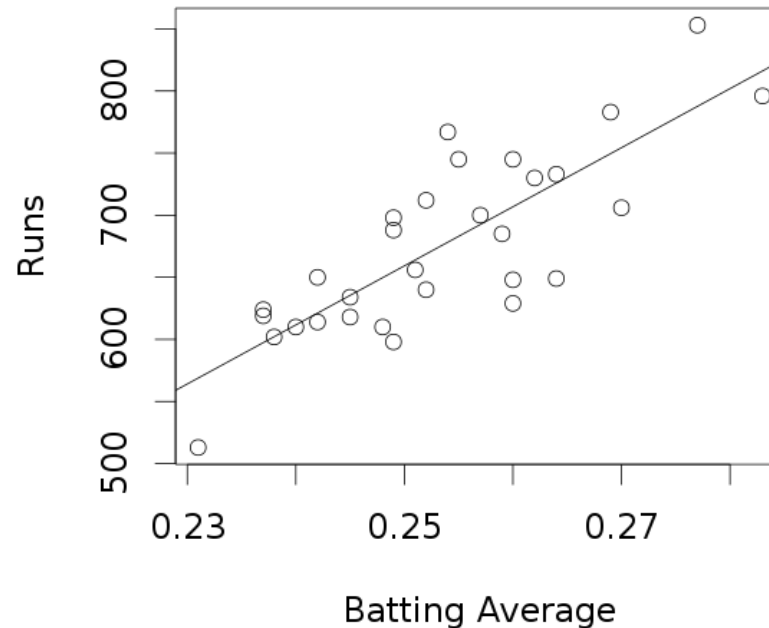


We can measure how well the line fits the data using the equation:

$$MSE = \frac{1}{n} \sum_i^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error

The square root of the MSE (RMSE) gives a sense of how much a points typically differ from the regression line



RMSE for predicting runs from BA is 51.35

- i.e., predictions typically off by 51 runs

Compare batting measures based on RMSE

We can also find the 'best' statistic based on the root mean squared error (RMSE)

- i.e., which statistic leads to a model with the minimal squared residuals

On lab 8 you will also calculate the RMSE between the number of runs a team scored and team level statistics for HR, BA, SLG, OBP, and OPS

Whichever statistic has the highest smallest RMSE we will say is the best statistics for capturing batting ability

We can then compare this to the best statistic found using correlation



Finding the least squares regression line

The least squares regression line

As we discussed, we can build a linear model using the statsmodels syntax:

```
lm = smf.ols('y ~ x', data = my_df).fit()
```

We can extract $\mathbf{b}_0$ (the intercept) and $\mathbf{b}_1$ (the slope) from the model using:

```
coefficients = lm.params
```

How does Python `.fit()` method finding the coefficients $\mathbf{b}_0$ and $\mathbf{b}_1$?

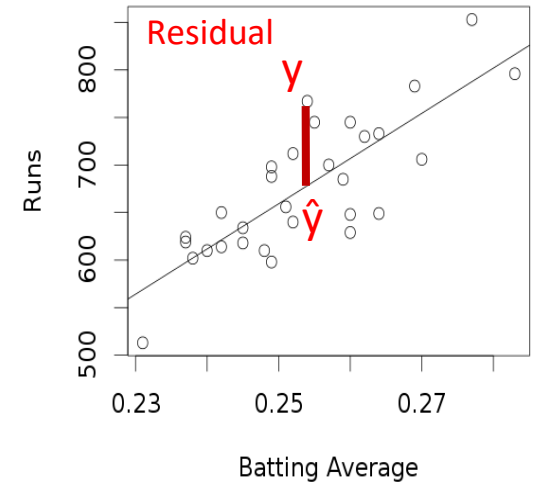
Least squares line

The **least squares line**, also called “**the line of best fit**”, is the line which minimizes the sum of squared residuals

i.e., the least squares line are the coefficients b_0 , and b_1 that minimize the Mean Squared Error (MSE)

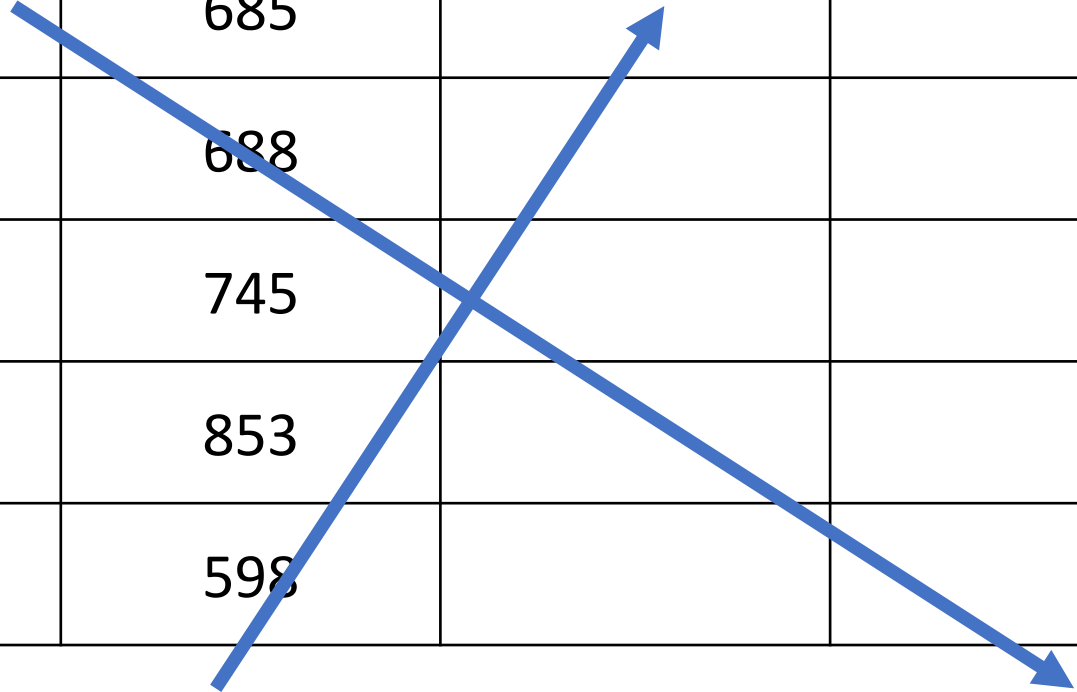
$$MSE = \frac{1}{n} \sum_i^n (y_i - \hat{y}_i)^2$$

You can experiment with this [using this app](#)



Residuals for the runs as a function of batting average

	Obs x (BA)	Runs obs (y) <i>(r in this case)</i>	Runs pred ($\hat{y}$) <i>($\hat{r}$ in this case)</i>	Residuals ($y - \hat{y}$) <i>($r - \hat{r}$ in this case)</i>
ARI	.259	685		
ATL	.249	688		
BAL	.260	745		
BOS	.277	853		
CHA	.249	598		


$$\hat{r} = -529.8 + 4755.7 \cdot BA$$

Residuals for the runs as a function of batting average

	Obs x (BA)	Runs obs (y) <i>(r in this case)</i>	Runs pred ($\hat{y}$) <i>($\hat{r}$ in this case)</i>	Residuals ($y - \hat{y}$) <i>($r - \hat{r}$ in this case)</i>
ARI	.259	685	702.0	
ATL	.249	688		
BAL	.260	745		
BOS	.277	853		
CHA	.249	598		

$$\hat{r} = -529.8 + 4755.7 \cdot BA$$

Residuals for the runs as a function of batting average

	Obs x (BA)	Runs obs (y) <i>(r in this case)</i>	Runs pred ($\hat{y}$) <i>($\hat{r}$ in this case)</i>	Residuals ($y - \hat{y}$) <i>($r - \hat{r}$ in this case)</i>
ARI	.259	685	702.0	-17.0
ATL	.249	688		
BAL	.260	745		
BOS	.277	853		
CHA	.249	598		

$$\hat{r} = -529.8 + 4755.7 \cdot BA$$

Residuals for the runs as a function of batting average

	Obs x (BA)	Runs obs (y) <i>(r in this case)</i>	Runs pred ($\hat{y}$) <i>($\hat{r}$ in this case)</i>	Residuals ($y - \hat{y}$) <i>($r - \hat{r}$ in this case)</i>
ARI	.259	685	702.0	-17.0
ATL	.249	688	654.4	33.6
BAL	.260	745	706.7	38.3
BOS	.277	853	787.6	65.4
CHA	.249	598	654.4	-56.4

$$\hat{r} = -529.8 + 4755.7 \cdot BA$$

Sum of squared residuals

	Runs obs (y)	Runs pred ($\hat{y}$)	Residuals (y - $\hat{y}$)	Residuals² (y - $\hat{y}$)²
ARI	685.0	702.0	-17.0	287.5
ATL	688.0	654.4	33.6	1129.0
BAL	745.0	706.7	38.3	1465.9
BOS	853.0	787.6	65.4	4282.4
CHA	598.0	654.4	-56.4	3181.0

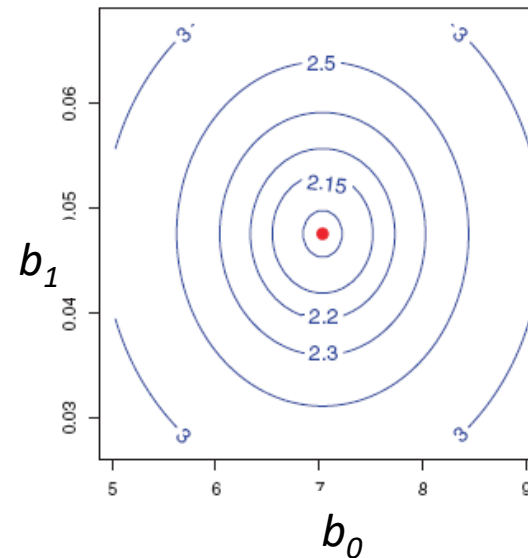
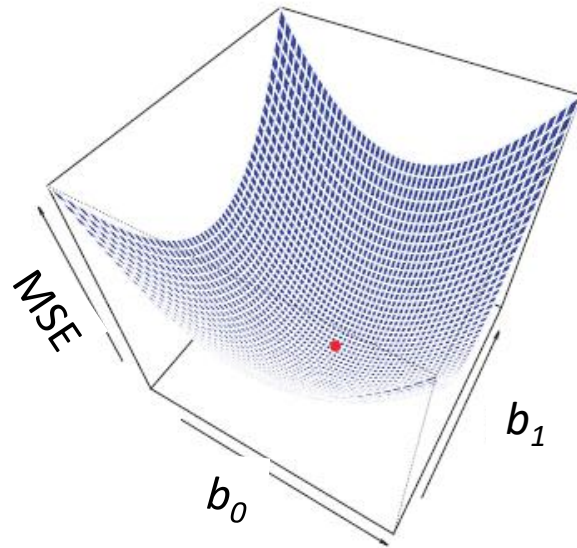
Taking the average of these deviations yields the Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_i^n (y_i - \hat{y}_i)^2$$

Finding the coefficients b_0 and b_1

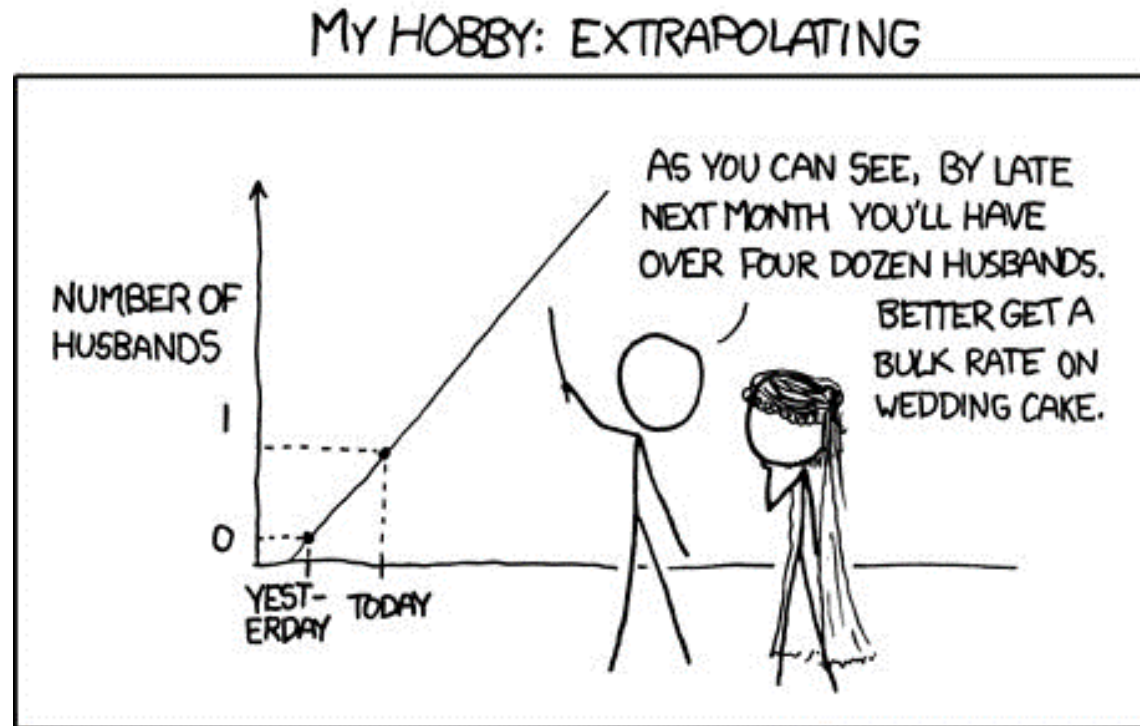
The regression coefficients can be found using calculus:

- Find b_0 and b_1 to minimize the mean squared error (MSE)
- This can be done by setting the partial derivative of the MSE with respect to b_0 and b_1 to 0 and solving for b_0 and b_1



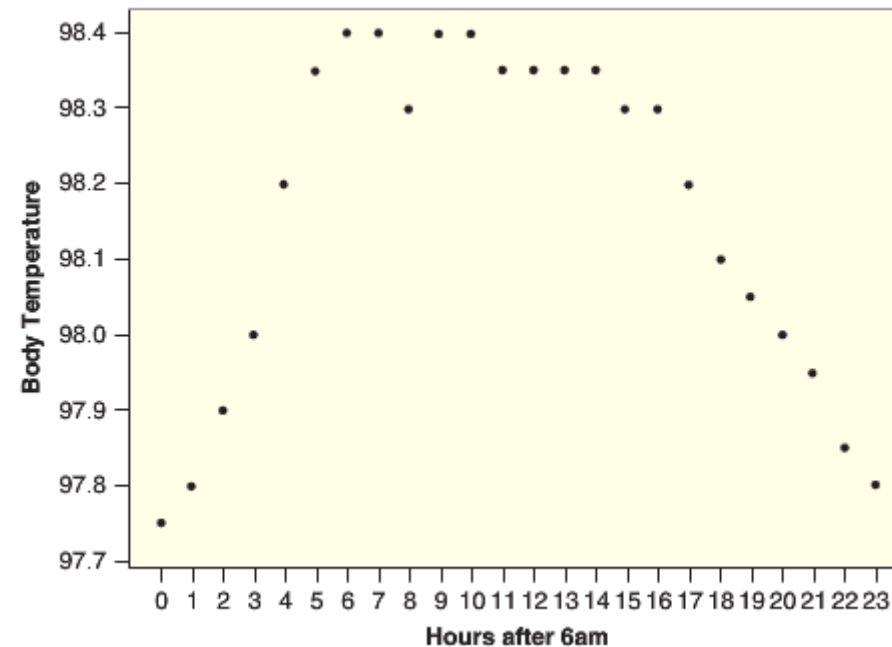
Regression caution # 1

Avoid trying to apply the regression line to predict values far from those that were used to create the line. i.e., do not extrapolate too far



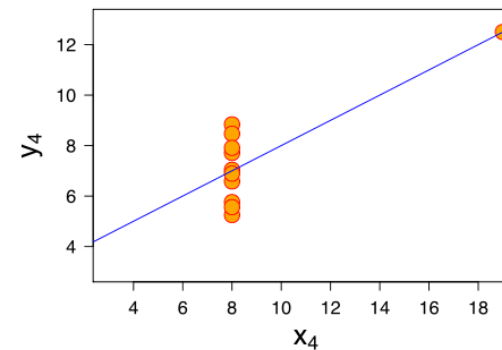
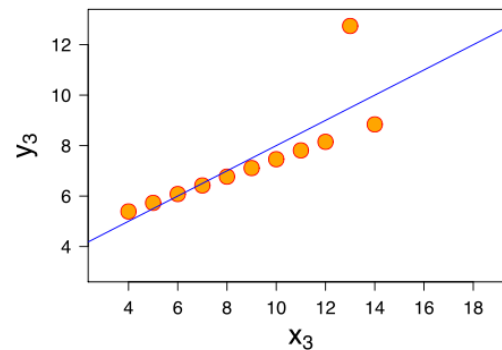
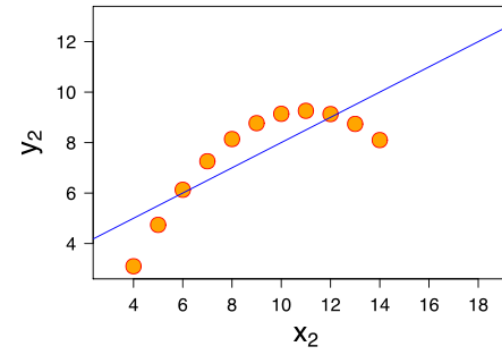
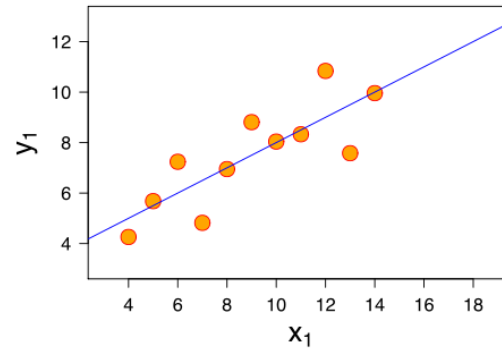
Regression caution # 2

Plot the data! Regression lines are only appropriate when there is a linear trend in the data



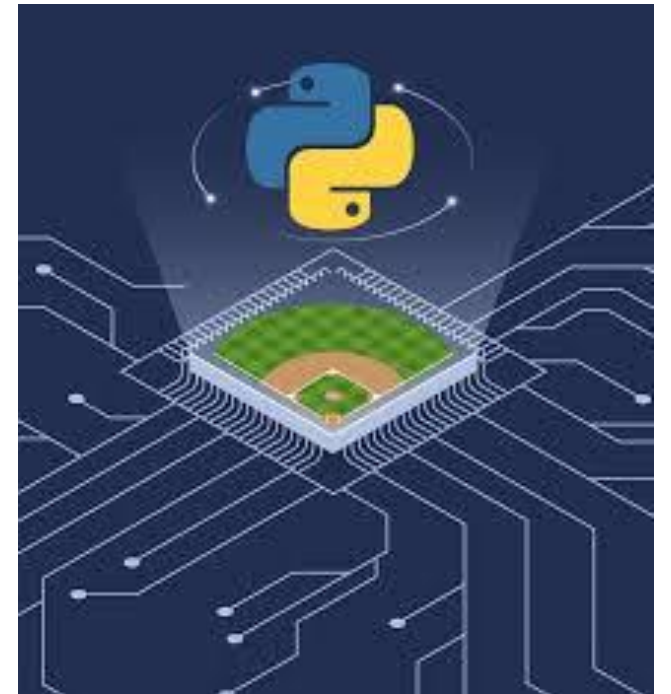
Regression caution #3

Be aware of outliers – they can have an huge effect on the regression line!



Residuals and the RMSE in Python

Let's quickly explore residuals and the RMSE in Python



Back to: who is better Ortiz vs. Jeter?



Derek Jeter



David Ortiz

Creating better 'metrics'

On lab 8 parts 1 and 2, you will try to determine which statistic between HR, BA, SLG, OBP, and OPS

But who says we can't do better than these statistics!



Creating better ‘metrics’

Batting average:

$$BA = [(1) \cdot \mathbf{1B} + (1) \cdot \mathbf{2B} + (1) \cdot \mathbf{3B} + (1) \cdot \mathbf{HR}] / \mathbf{AB}$$

Slugging percentage:

$$\text{Slug} = [(1) \cdot \mathbf{1B} + (2) \cdot \mathbf{2B} + (3) \cdot \mathbf{3B} + (4) \cdot \mathbf{HR}] / \mathbf{AB}$$

On-base percentage:

$$OBP = [(1) \cdot \mathbf{BB} + (1) \cdot \mathbf{HBP} + (1) \cdot \mathbf{1B} + (1) \cdot \mathbf{2B} + (1) \cdot \mathbf{3B} + (1) \cdot \mathbf{HR}] / \mathbf{PA}$$

Optimal statistic:

$$OPT = b_1 \cdot \mathbf{BB} + b_2 \cdot \mathbf{HBP} + b_3 \cdot \mathbf{1B} + b_4 \cdot \mathbf{2B} + b_5 \cdot \mathbf{3B} + b_6 \cdot \mathbf{HR} + b_0$$

We want to find the “best” b_i ’s for predicting how many runs a team scored

What are the optimal weights?

Any ideas for the best b_i 's ?

$$\text{OPT} = b_1 \cdot \text{BB} + b_2 \cdot \text{HBP} + b_3 \cdot \text{1B} + b_4 \cdot \text{2B} + b_5 \cdot \text{3B} + b_6 \cdot \text{HR} + b_0$$

We can use multiple regression, to find these optimal b_i 's !

Multiple regression

In multiple regression we try to predict a quantitative response variable y using several predictor variables $x_1, x_2, \dots, x_k$

For multiple linear regression our equation has the form of:

$$\hat{y} = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k$$

Multiple regression

$$\hat{y} = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k$$

There are many uses for multiple regression models including:

- To make predictions as accurately as possible
- To understand which predictor variables are related to the response variable
- To create new statistics (“metrics”) that give a useful numerical description of a phenomenon



What are the optimal weights?

Any ideas for the best b_i 's ?

$$\text{OPT} = b_1 \cdot \text{BB} + b_2 \cdot \text{HBP} + b_3 \cdot \text{1B} + b_4 \cdot \text{2B} + b_5 \cdot \text{3B} + b_6 \cdot \text{HR} + b_0$$

For example, we could use multiple regression to find the b_i 's that minimize sum of $(R - \text{OPT})^2$

```
lm = smf.ols('R ~ BB + HBP + H + X2B + X3B + HR', data = teams_162).fit()
```

```
the_params = lm.params
```

How low can you go?

Can you come create additional variables that will lead to an even better RMSE?

Lab 8!



Multiple regression in Python

Let's fit a multiple regression model that predicts:

- The number of wins a team has in a season (y)
- Based on the number of runs the score (R) and the number of runs they allow (RA)

